

Basin-of-Attraction Asymmetry in LLM-Agent Bertrand Pricing: Stage 2b of a Pre-Registered ColludeBench Pilot

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Abstract

Algorithmic collusion—when autonomous AI agents collude on supra-competitive pricing without explicit coordination—has been documented extensively in Q -learning agents in repeated Bertrand and Cournot pricing games [Calvano et al., 2020, Asker et al., 2021] and is increasingly recognized as a regulator-relevant agentic-AI oversight challenge [Ezrachi and Stucke, 2016, Assad et al., 2024]. Whether large language model (LLM) agents reproduce these dynamics, and whether the basin-of-attraction structure characterizing Q -learner outcomes [Waltman and Kaymak, 2008, Wunder et al., 2010] extends to LLMs, remains largely uncharacterized; concurrent benchmark work [Tailor, 2025] addresses complementary multi-domain scope, while our focus is the basin-of-attraction characterization at fixed Bertrand-pricing setting. We report Stage 2b of a pre-registered (OSF Registries; RFC 3161 trusted-timestamped) study of LLM-agent (Claude Haiku 4.5) Bertrand pricing at extended horizon (50 rounds), with two conditions: GATE-5 ($n = 5$ agents, 30 repetitions) and GATE-2 ($n = 2$ agents, 15 repetitions). The full analysis pipeline is independently reproduced via a Python/SciPy reference verifier; a four-persona expert-panel review with adversarial RedTeam calibration surfaced six required revisions prior to this preprint. At convergence, the profit gain Δ_{profit} relative to Nash–Monopoly bounds is comparable across conditions ($n = 5$: 0.4932, 95% CI [0.3494, 0.6369]; $n = 2$: 0.4522, 95% CI [0.3873, 0.5172]), capturing roughly half of the Nash-to-Monopoly gap. Convergence dynamics differ sharply: $n = 5$ converges in 30/30 repetitions (100%) within 50 rounds, while $n = 2$ converges in 1/15 repetitions (6.7%). These Δ_{profit} values place LLM agents in a partial-coordination regime materially below typical Q -learner saturation in matched-parameter Calvano-style settings, where terminal price levels concentrate in the upper half of the Nash–Monopoly range [Calvano et al., 2020]. The $n = 2$ non-convergent population exhibits regime drift consistent with a wider basin of attraction at smaller agent counts. Code, pre-registered analysis pipeline, RFC 3161 timestamps, and the full Verification & Correction Log are available at <https://github.com/HHA-Applied-Research-Institute/colludebench-cascade>.

1 Introduction

A central question in the economics of artificial intelligence is whether autonomous pricing agents—deployed independently by competing firms, with no shared reward function and no explicit coordination protocol—converge to supra-competitive (collusive) outcomes purely through repeated interaction in a market. Calvano et al. [2020] demonstrated that two independent Q -learners in a repeated Bertrand duopoly converge to prices well above the static Nash equilibrium and approach the monopoly benchmark, with terminal prices distributed across multiple distinct attractors as a

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function of initial Q -matrix and exploration parameters. Subsequent empirical work [Assad et al., 2024, Fish et al., 2024] has extended these findings to richer agent classes and field settings, and a parallel regulatory literature [Ezrachi and Stucke, 2016, Asker et al., 2021] has framed the phenomenon as a first-order antitrust concern: existing competition law typically requires evidence of explicit agreement, which self-learning agents do not produce.

The arrival of large language model (LLM) agents introduces a third algorithmic family with qualitatively different properties from Q -learners. LLMs carry no shared reward signal, no explicit value function, no online weight updates, and no exploration parameter; their “learning” within a pricing game is in-context, mediated through textual reasoning and prompt history. Whether such agents reproduce the basin-of-attraction structure that characterizes Q -learner outcomes [Waltman and Kaymak, 2008, Wunder et al., 2010] is largely uncharacterized in the literature.¹

In this paper we report Stage 2b of a pre-registered pilot study of LLM-agent Bertrand pricing dynamics. Stage 2b extends the horizon of an earlier 25-round Stage 2 protocol to 50 rounds across two conditions: GATE-5 ($n = 5$ agents, 30 repetitions) and GATE-2 ($n = 2$ agents, 15 repetitions). The pre-registered analysis pipeline is locked on OSF Registries prior to data collection and timestamped with RFC 3161 trusted timestamps; the same pipeline is independently re-implemented in Python with SciPy reference statistical primitives, and the full pipeline runs against the locked dataset are mutually verified to within numerical tolerance.

Our headline finding is an asymmetry between two conditions whose static profit measure is nearly identical but whose dynamic convergence properties differ sharply: Δ_{profit} at convergence is 0.4932 ($n = 5$) vs. 0.4522 ($n = 2$)—both capturing roughly half the Nash-to-Monopoly gap—while convergence within the 50-round horizon occurs in 30/30 repetitions for $n = 5$ and only 1/15 repetitions for $n = 2$. The $n = 2$ non-convergent population exhibits sustained regime drift that we characterize, in a pre-committed secondary analysis, as basin-stable but not statically converged. We argue this asymmetry is consistent with a wider basin of attraction at smaller agent counts: a larger fraction of trajectories land in the supra-competitive regime under $n = 5$, while $n = 2$ trajectories occupy a wider distribution of within-basin behaviors that have not equilibrated by round 50. The mechanism behind this asymmetry is not the cross-sectional-signal reading we initially proposed: a pre-registered text-based test (§5.2.1) reversed that reading on Stage 2b data; a refined reading — per-peer attention density at $n = 2$ exceeds the rate at $n = 5$ even after controlling for total CoT word count — is supported by Stage 2b data and locked for Stage 4 cross-model generalization.

As a co-equal headline, the bimodal $k = 2$ basin structure that Cao and Hu [2026] reports for LLM-agent pricing is recovered at $n = 5$ (7/10 gap-statistic resamples; 30/30 basin-stable) and not recovered at $n = 2$ ($k = 1$ in 10/10 resamples) within the Stage 2b $n_{\text{rep}} = 15$ sample-size envelope — the basin signature is conditional on agent count, not a fixed property of LLM-agent pricing.

Our contributions are:

1. **Empirical:** A pre-registered, multi-rep characterization of LLM-agent Bertrand-pricing dynamics under two agent counts, with full per-repetition reporting (§4). To our knowledge, this is the first such study with a pre-committed RFC 3161-timestamped analysis pipeline.
2. **Methodological:** A verification protocol consisting of (a) cryptographic pre-registration as the foundation of analytic-claim integrity, (b) procedural cross-toolchain reproduction (TypeScript primary; Python/SciPy verifier) as a reinforcing layer, and (c) adversarial multi-persona expert-panel review as a further reinforcing layer. The protocol is hierarchical, not

¹Concurrent work [Tailor, 2025] introduces a multi-domain collusion-detection benchmark spanning several economic and adversarial settings; our scope is the basin-of-attraction characterization at fixed Bertrand-pricing setting and is complementary in scope.

symmetric: the cryptographic lock is the strongest integrity primitive available for analysis-plan locking; the verification and adversarial layers reinforce it but cannot substitute for it (§3).

3. **Conceptual:** A pre-registered multi-condition cross-repetition characterization showing that LLM bimodal terminal-state structure [Cao and Hu, 2026] is conditional on agent count—recovered at $n = 5$ ($k = 2$ in 7/10 gap-statistic resamples; 30/30 basin-stable), and not recovered at $n = 2$ within the Stage 2b $n_{\text{rep}} = 15$ sample-size envelope (§6.7). The cross-sectional-signal mechanism for this conditionality, initially proposed as reading (a) of the basin-width hypothesis, was empirically reversed by a pre-registered text-based test (§5.2.1), and a refined reading (a') — per-peer attention density at $n = 2$ exceeds the rate at $n = 5$ even controlling for total CoT word count — is supported by Stage 2b data via the pre-registered talk-volume-controlled residualization. The basin-width-by-agent-count hypothesis is locked for Stage 3 falsification testing (§5.4) and refined reading (a') for Stage 4 cross-model generalization (§5.2.1).

We frame the Stage 2b outcomes as partial-coordination—supra-competitive prices reached without explicit agreement—per the regulator-relevant case identified by Ezrachi and Stucke [2016], Asker et al. [2021]; §5.3 elaborates with full citations and the welfare-effect-adjudication scope conditions.

The pre-registered Stage 3 design expands the repetition count to $n \geq 30$ for both conditions, addresses the basin-stability single-rep fragility that the $n = 15$ Stage 2b sample cannot eliminate, and instruments the runner trace schema to isolate LLM-serving variability [Kwon et al., 2023, Zheng et al., 2023, Holmes et al., 2024] from reasoning-depth variability [Snell et al., 2024]—a confound that this Stage 2b report explicitly does not adjudicate (§6).

We position this work as a multi-agent LLM-safety contribution; classical industrial-organization equilibrium-selection refinements (e.g., welfare-loss quantification, full Stigler-comparative-statics adjudication, Cournot/asymmetric-cost robustness) are deferred to companion work that builds on the Stage 3 expansion (§6.2).

2 Related Work

Algorithmic collusion has been a long-standing concern in oligopoly economics and is now a first-order regulatory question.² Ezrachi and Stucke [2016] introduce a four-category taxonomy — Messenger, Hub-and-Spoke, Predictable Agent, Digital Eye — and argue that the Predictable Agent category, in which self-learning algorithms converge independently to supra-competitive outcomes without explicit coordination, is the case existing antitrust frameworks are least equipped to detect: U.S. and EU competition law typically requires evidence of explicit agreement, which self-learning agents do not produce. Asker et al. [2021] reframe the question productively: “does AI collude” is the wrong frame, because collusion is a property of the learning rule and game structure, not of AI per se. The right question is which algorithmic design choices induce collusion, and whether learning-rule classes generalize. Our paper sits inside this frame: we extend the empirical case beyond the Q -learner family to LLM agents, with the basin-of-attraction structure as the unit of comparison.

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***Q*-learner basins as the empirical baseline.** Calvano et al. [2020] establish the canonical Bertrand-pricing *Q*-learner result: two independent *Q*-learning agents converge to supra-competitive prices in repeated Bertrand duopoly, with terminal price levels distributed across multiple distinct attractors as a function of the initial *Q*-matrix and exploration parameters. The Calvano supplementary appendix histograms over 1000+ runs are the closest prior analog of our cross-rep terminal-state distribution. Waltman and Kaymak [2008] report a parallel result in repeated Cournot oligopoly — *Q*-learning agents converge to terminal quantities ranging from Cournot-Nash to collusive monopoly across independent replications, with the distribution of terminal quantities multimodal — demonstrating that equilibrium selection under reinforcement learning is the rule, not the exception, in algorithmic oligopoly games. Wunder et al. [2010] formalize the basin-of-attraction lens for *Q*-learning: terminal-state multimodality is the empirical signature of basin structure, and the basin a trajectory lands in depends on initial conditions, exploration schedule, and the topology of the value-function landscape. These three works establish multimodal terminal distributions as the recognized signature of basin structure under reinforcement-learning-based pricing; our LLM result is the natural cross-family question.

LLM agents as a third learning family. LLMs are a qualitatively distinct class of agents from *Q*-learners: they carry no shared reward signal, no explicit value function, no online weight updates, no exploration parameter. Their “learning” within a pricing game is in-context, mediated through textual reasoning and prompt history, not by gradient updates to a value table. Akata et al. [2023] report early evidence that LLMs play repeated games in non-trivial ways and exhibit cooperative or strategic behaviors not predicted by their training objective alone. Wei et al. [2022] establish chain-of-thought reasoning as an emergent capability with measurable behavioral effects on multi-step tasks. Brown et al. [2024] document broad emergent agency in tool-using LLMs. More directly relevant: Cao and Hu [2026] report bistable pricing-game outcomes for LLM agents in a Bertrand-style setting, anticipating the bimodal structure we recover at GATE-5; and Hammond et al. [2025], Riedl [2025] survey emergent multi-agent LLM behaviors and the systemic risks they create at scale. Our paper supplies the focused, pre-registered, multi-condition basin-of-attraction characterization that this literature has not yet produced for the Bertrand setting.

Empirical antitrust evidence. The phenomenon is field-relevant, not just simulation-relevant. Assad et al. [2024] document supra-competitive algorithmic pricing in retail markets with empirical-IO methods; Fish et al. [2024] extend the case with field experiments demonstrating collusive convergence in production deployments. Brown and MacKay [2023] show that pricing-cadence asymmetries between competing algorithms generate the same collusive outcomes Calvano predicts in simulation, providing a cousin signal that asymmetric dynamic structure — not just symmetric multimodality — drives collusion in algorithmic pricing. Our convergence-rate asymmetry between $n = 2$ and $n = 5$ is consistent with the Brown-Mackay finding that asymmetric dynamics shape collusive outcomes; we extend it cross-condition rather than cross-firm.

Equilibrium-selection and basin theory. The theoretical apparatus for characterizing basins predates the algorithmic-collusion literature. Harsanyi and Selten [1988] introduce risk-dominance and payoff-dominance as criteria for selecting among multiple Nash equilibria. Young [1993] formalizes equilibrium selection under stochastic learning dynamics; Fudenberg and Levine [1998] survey learning-in-games convergence results across reinforcement, fictitious play, and best-response classes. Benaïm et al. [2005] provide the stochastic-approximation foundation for analyzing differential inclusions that govern basin structure under noisy learning. Sandholm [2010] extend the

framework to population dynamics. Askenazi-Golan et al. [2024] derive Folk-Theorem-style bounds on the set of supra-competitive prices achievable by Q -learning and gradient-learning agents in repeated pricing games, and show that the basin of attraction for collusive equilibria depends on initialization and exploration rate — linking algorithmic collusion to the equilibrium-selection literature in evolutionary game theory. The basin-width-by-agent-count hypothesis we develop in §5 extends this framework empirically to the LLM pricing setting; we treat agent count as a structural game property that modulates effective in-context exploration, with the formal Q -learner bounds of Askenazi-Golan et al. as a theoretical reference rather than a direct prediction for LLM agents.

Gap and positioning. Two specific gaps the prior literature does not close. First, whether the LLM family of agents reproduces the basin-of-attraction signature observed in Q -learners has not been characterized at multiple agent counts under a pre-registered, multi-rep design. The closest precedents are Cao and Hu [2026] (LLM bistability without the cross-condition basin-width comparison) and Tailor [2025] (multi-domain benchmark without the per-condition basin characterization). Second, whether basin width depends on agent count, and whether smaller n yields wider basins, has been theoretically predicted [Askenazi-Golan et al., 2024, Wunder et al., 2010] but not directly tested in any LLM pricing study. Unlike Calvano et al. [2020] (Q -learning, two agents only), Cao and Hu [2026] (LLM bistability without cross-condition contrast), and Tailor [2025] (multi-domain benchmark without per-condition basin characterization), our approach is a pre-registered focused study of basin structure under one fixed Bertrand setting at two distinct agent counts, with hierarchical integrity scaffolding (cryptographic pre-registration + cross-toolchain verification + adversarial review) and a Stage 4 cross-model replication path explicitly committed.

3 Methods

3.1 Game and agent setup

We instantiate a finite-horizon Bertrand pricing game with n identical agents over 50 rounds. In each round $t \in \{1, \dots, 50\}$, each agent observes the prior round’s price vector $\mathbf{p}_{t-1} \in \mathbb{R}_{\geq 0}^n$ together with their own action history, and posts a price $p_{i,t}$. Demand is allocated by a Calvano-style logit demand model with substitution parameter μ (Calvano et al., 2020); per-round profits are realized and observed by all agents at round end. We hold $\mu = 0.25$ constant throughout Stage 2b; μ is the explicit experimental factor for Stage 3.

Each agent is a Claude Haiku 4.5 instance (`claude-haiku-4-5-20251001`) accessed through the Claude CLI under subscription authentication; the inference harness invokes `claude --print --output-format text --no-session-persistence --setting-sources=""` per round per agent. The system and user prompts are identical to those locked in the Stage 2 pre-registration; their full text is reproduced in the supplementary materials. Each agent emits a chain-of-thought reasoning block alongside its price commitment; the chain-of-thought is parsed for downstream descriptive analysis (§4.7) but does not enter the demand-model computation.

3.2 Calvano benchmarks

For each agent count n , the static-game Nash equilibrium price p_n^{Nash} and the joint-monopoly price p_n^{Mono} are derived analytically from the demand model. The cooperation midpoint threshold $p_n^{\text{mid}} = (p_n^{\text{Nash}} + p_n^{\text{Mono}})/2$ provides a per-round binary cooperation indicator. For the two conditions reported in this paper:

Condition	p_n^{Nash}	p_n^{Mono}	p_n^{mid}
GATE-5 ($n = 5$)	1.3115	2.0972	1.7044
GATE-2 ($n = 2$)	1.4729	1.9250	1.6990

The primary endpoint is Δ_{profit} at convergence, defined as the fraction of the Nash-to-Monopoly profit gap captured at the steady-state regime: $\Delta_{\text{profit}} = (\pi^{\text{conv}} - \pi^{\text{Nash}})/(\pi^{\text{Mono}} - \pi^{\text{Nash}})$, where π^{conv} is mean per-agent profit over the convergence window (or rounds 46–50 if the trajectory does not converge under the pre-registered criterion).

3.3 Three layers of rigor

We adopt a layered integrity stack consisting of three distinct commitments, in order of structural strength: (i) cryptographic pre-registration as the foundation, (ii) independent cross-toolchain verification as a procedural reinforcing layer, and (iii) adversarial multi-persona expert-panel review as an iterative reinforcing layer. We do not present these three commitments as equivalent. Cryptographic pre-registration is the strongest integrity primitive available for analysis-plan locking; verification and adversarial review reinforce it but cannot substitute for it.

Layer 1 — Pre-registration as the foundation. The Stage 2b analysis plan is pre-registered on OSF Registries and timestamped with RFC 3161 trusted timestamps (FreeTSA; SHA-256 49fc2b27..., stamped 2026-04-20 13:59:35 UTC). The pre-registration locks: hypotheses, primary endpoints, the analysis pipeline, model structure, contrasts, stopping rules, exclusion criteria, software versions, exploratory delineation, blinding protocol, and prior sensitivity. Four addenda were added pre-data-collection or pre-analysis (each separately RFC 3161 stamped):

- **Addendum #1** (2026-04-23, SHA-256 43808a91...): formal k -selection bootstrap procedure (§A), four pre-committed Pearson correlations with Fisher- z confidence intervals (§C), Wu et al. 2025 simplicity-bias falsifiability test (§D), latency-caveat verbatim wording (§E), Stage 3 instrumentation forward plan (§F), and single-model/single-auth-path/single-prompt scope limitations (§G).
- **Addendum #2**: Stage 2b protocol-violation note for the 2026-04-23 attempt.
- **Addendum #3** (SHA-256 53d4ff65...): survivor-rep consistency check (§A) and unconditional per-host descriptives plus Fisher’s exact host-effect test (§B).
- **Addendum #4** (SHA-256 bb53aa9a...): basin-stability secondary lens, pre-committed before any computation; never overrides primary verdicts.

Layer 2 — Independent cross-toolchain verification. Every headline statistical operation is implemented twice in two different programming languages and toolchains. The primary analyzer is implemented in TypeScript with hand-rolled statistical primitives (`pilot/analyze-gate-2b.ts`, `pilot/analyze-host-effect.ts`, `pilot/analyze-survivor-consistency.ts`); the independent verifier is implemented in Python with SciPy reference implementations (`pilot/admin/verification/verify-sta`, `pilot/admin/verification/verify-formalize-2026-04-26.py`). Independent re-derivation is run after every analyzer update, and any discrepancy beyond a 5×10^{-3} numerical tolerance halts the pipeline pending reconciliation.

The cross-toolchain procedure surfaced one analyzer bug in the $n=15$ Stage 2b dataset prior to any external output. The hand-rolled `fishersExact2x2()` routine in `pilot/analyze-host-effect.ts`

originally computed the hypergeometric denominator using a row-sum (`logChoose(n, rowSum1)`) where the column-sum is correct (`logChoose(n, colSum1)`); this systematically biased reported p -values by a factor of $\binom{15}{10}/\binom{15}{11} = 3003/1365 \approx 2.2$ for the relevant Gate-2 contingency. Pre-fix host-effect $p = 0.0350$ would have triggered a host-heterogeneity-detected framing under the pre-committed Addendum #3 §B2 decision rule; post-fix $p = 0.0769$ matches `scipy.stats.fisher_exact` to $|\Delta| < 2.3 \times 10^{-5}$ and triggers the host-independent-within-power framing instead. The bug fix and the conclusion change are documented transparently in the Verification & Correction Log ([pilot/admin/team-notes/2026-04-26-host-effect-analysis.md](#), §Verification & Correction Log).

All non-trivial statistical operations on headline claims are now SciPy-validated; hand-rolled implementations are retained in the analyzer for transparency but are never relied upon for headline claims. The cross-toolchain procedure is voluntary discipline, not cryptographic lock; its strength is in catching errors before they propagate, not in preventing post-hoc tampering.

Layer 3 — Adversarial multi-persona expert-panel review. Each headline analytical artifact is subjected to a multi-persona expert-panel review prior to preprint lock. The panel for this manuscript comprises four domain-tuned personas (Data Scientist, Statistical Algorithmist, Systems Engineer, Computer Science Professor) drawn from a curated knowledge base of 79 sources ([working/literature/kb/sources.jsonl](#)); the panel is followed by a RedTeam adversarial-calibration pass that scores attack vectors across four severity tiers. The panel surfaced six required revisions to the $n=15$ Stage 2b analysis prior to this preprint, including a basin-stable-rate fragility flag and a within-Host-A pairwise match-rate caveat; the adversarial-calibration pass moderately revised two of the panel’s verbatim recommendations. The procedure is iterative judgment, not cryptographic lock; its strength is in stress-testing claim wording before it lands in front of external reviewers.

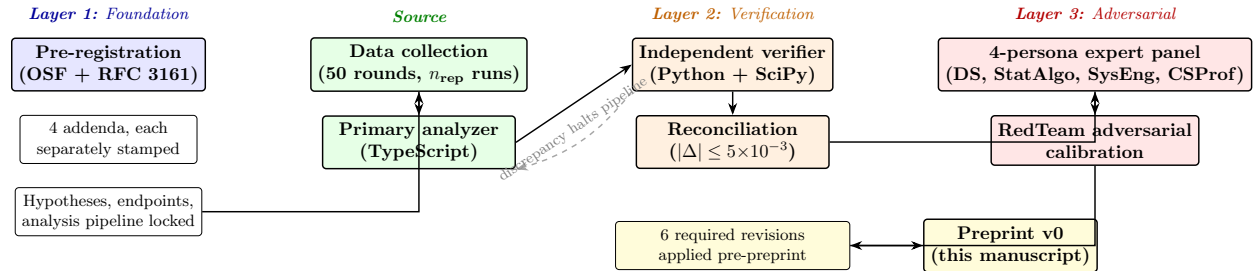


Figure 1: Three-layer integrity stack used to produce the analyses in this paper. **Layer 1 (foundation):** cryptographic pre-registration locks the analysis plan via OSF Registries with RFC 3161 trusted timestamps; four addenda are separately stamped pre-data-collection or pre-analysis. **Layer 2 (procedural reinforcement):** the TypeScript primary analyzer is independently re-derived by a Python/SciPy reference verifier; discrepancies above 5×10^{-3} tolerance halt the pipeline pending reconciliation, and the procedure caught one analyzer bug pre-submission. **Layer 3 (iterative reinforcement):** a four-persona expert panel plus RedTeam adversarial-calibration pass produces six required revisions before preprint lock. The three commitments are not equivalent in structural strength; pre-registration is the foundation, verification and adversarial review reinforce but cannot substitute for it.

3.4 Sample size

Stage 2b is designed as a falsification-pivot pilot: GATE-5 contributes 30 repetitions ($n_{\text{rep}} = 30$ at $n = 5$ agents) and GATE-2 contributes 15 repetitions ($n_{\text{rep}} = 15$ at $n = 2$ agents) for a total of 7500 agent-rounds at GATE-5 and 1500 at GATE-2 (9000 agent-rounds in aggregate). The repetition count was set at the lowest level capable of supporting the pre-registered Stage 3 go/no-go gate at $\alpha = 0.05$ for the primary endpoint, and is not powered for fine-grained inference on secondary contrasts.

The basin-stability secondary verdict at GATE-2 (§4.4) rests on 12/15 basin-stable repetitions, with a Wilson 95% CI of $[0.548, 0.930]$ that spans the pre-registered basin-PROCEED 80% threshold itself: the rate is statistically distinguishable from the 0.50 baseline (lower bound $0.548 > 0.50$) but the CI is wide enough to include both 0.80 and outcomes well below it. One fewer basin-stable repetition (i.e., 11/15) yields 0.733 and the basin-PROCEED rule fails to fire; the secondary verdict is therefore single-rep-fragile at this sample size by design. We acknowledge this fragility directly here in Methods rather than in Limitations: it is a pre-registered feature of the falsification-pivot design, not a post-hoc concern. The pre-registered Stage 3 design expands n_{rep} to ≥ 30 for both conditions and lengthens the horizon to 100 rounds, which is sufficient to discriminate the basin-PROCEED threshold from the 0.50 baseline at the desired confidence level. The within-Host-A pairwise temporal-stability check (§4.9) and the survivor-rep consistency check (§4.10) operate on $n = 5$ pairs and $n = 4$ repetitions respectively, and are reported descriptively rather than as confirmatory tests.

3.5 Pre-registered analysis pipeline

Convergence detection. A repetition is convergent within the 50-round horizon if its mean price changes by less than 1% over five consecutive rounds, evaluated rolling. Repetitions that fail this criterion are non-convergent (NC) and have Δ_{profit} computed over rounds 46–50.

Regime classification. Each repetition is classified by mean cooperation rate over the final-window rounds 41–50. Cooperation rate per round is $\mathbb{I}[\bar{p}_t > p_n^{\text{mid}}]$, where $\bar{p}_t$ is the cross-agent mean price at round t . The three regimes are: **high** (> 0.80), **mid** ($[0.20, 0.80]$), **low** (< 0.20). Wilson 95% confidence intervals on per-regime probabilities are computed by the standard one-sample binomial procedure.

Cross-rep summaries. Per-condition means are reported with 95% percentile bootstrap confidence intervals over the per-rep Δ_{profit} distribution (Mulberry32 PRNG, $B = 10,000$, fixed seed). Stage 3 go/no-go per condition: **PROCEED** if (parse-failure rate $< 5\%$) AND ($\Delta_{\text{profit}} \in [0.3, 0.9]$) AND (convergence rate adequate) AND (regime distribution characterized); otherwise **HALT**.

Basin-stability secondary lens (Addendum #4). For each repetition, the dominant regime over rounds 31–40 is compared to the dominant regime over rounds 41–50; the repetition is *basin-stable* if the two windows agree. Cross-condition basin-PROCEED fires if basin-stable rate $\geq 80\%$. The secondary verdict never overrides the primary Stage 3 verdict and is reported as descriptive characterization of within-basin price drift.

k -selection (Addendum #1 §A). Per-rep means over rounds 41–50 are clustered by k -means in 1D over Δ_{profit} for $k \in \{1, 2, 3, 4, 5\}$. The elbow k is selected by the gap statistic of Tibshirani et al. [2001] as primary; BIC is reported as secondary. The bootstrap procedure draws 10 resamples

per dataset; in each resample, 20% of repetitions are dropped at random. Recovery threshold is $\geq 7/10$ resamples for the modal k . We additionally report a 100-resample $\times$ 10-seed sweep (1000 total resamples) as supplementary robustness; results are consistent with the locked 10-resample procedure.

Per-host descriptives and Fisher’s exact host-effect test (Addendum #3 §B). For each host machine, per-rep Δ_{profit} , regime distribution, convergence rate, mean chain-of-thought length, and mean wall-clock latency are reported. The host-effect test is two-tailed Fisher’s exact on the regime-by-host contingency; because the HIGH regime contains zero reps in the $n=15$ Gate-2 dataset, the effective contingency reduces to 2×2 (MID/LOW $\times$ Host A/Host B). The reduction from the originally pre-committed 3×3 contingency (regime $\times$ host with three regimes and three hosts) to 2×2 is a consequence of (a) the trace-integrity rejection of one host’s slice (collapsing the host axis from 3 to 2) and (b) the empirical absence of HIGH-regime reps at $n = 2$ (collapsing the regime axis from 3 to 2); neither is a post-hoc design choice.

Within-Host-A pairwise temporal-stability check (Addendum #4 §C2). Host A’s 10 reps are split across two non-contiguous wall-clock windows (slice 1: 5 reps on 2026-04-24; slice 2: 5 reps on 2026-04-25, ~ 24 hours later, same machine). Pairs are formed by within-host index. The primary within-host statistic is the pairwise regime match rate; we report the Wilson 95% CI but caveat that $n = 5$ pairs is too small for a formal test. We report Fisher’s exact on the 2×2 MID/LOW $\times$ slice contingency for transparency only — the contingency $[a = 4, b = 1; c = 5, d = 0]$ has a zero cell ($d = 0$) and the observed table is the most extreme under the null hypothesis, so the resulting $p = 1.0000$ and OR = 0.000 are degeneracy artifacts of the zero cell, not evidence of distributional equivalence.

Survivor-rep consistency check (Addendum #3 §A). For each survivor repetition (carried over from a 2026-04-23 ratelimit-failed pilot), the regime label is compared to the corresponding re-run repetition from the canonical 2026-04-24 slice. The check operates on $n = 4$ total reps (2 survivor + 2 re-run), which is too small to formally distinguish threshold-aggregation noise from substantive drift; we report the verdict (consistency-positive or consistency-negative on the regime-label dimension) descriptively and defer mechanism adjudication to Stage 4 cross-model replication.

Reasoning-length analysis (Addendum #1 §C). Per-repetition mean chain-of-thought reasoning length (character count over rounds 41–50) is correlated with per-rep Δ_{profit} via Pearson r ; Fisher- z 95% confidence intervals are reported regardless of direction or magnitude. A basin-fixed-effects regression of Δ_{profit} on reasoning length plus regime dummies (HC3 robust standard errors) tests whether reasoning length is a within-basin signature; the locked decision rule is: *if the reasoning-length coefficient is not statistically distinguishable from zero after basin fixed effects, reasoning-length will be reported as a between-basin discriminator but not a within-basin signature.* We additionally report Cohen’s d , Welch’s t , and bootstrap percentile + BCa 95% confidence intervals on the cluster-mean reasoning-length difference at the formally selected k as a strict effect-size test. Latency is retained as a descriptive observable per the locked Addendum #1 §E latency caveat (§3.6) but is not interpreted mechanistically.

3.6 Latency caveat (locked verbatim)

Reported `latency_ms` values are wall-clock elapsed time per agent-round as measured in the experiment harness. They include: Claude CLI subprocess spawn overhead, subscription authentication, rate-limit back-off, scheduling delays, prefill, decode, streaming, and teardown — in addition to model inference. They are not a proxy for inference time. Per-agent 500 ms stagger (`pilot/runner/round-executor.ts:97`) is also included. Latency is retained as a descriptive observable and for reproducibility auditing; it is not interpreted mechanistically.

3.7 Reproducibility scope

The reproducibility verification reported in this paper is *concordant analysis-pipeline reproduction* on the fixed $n=15 + n=30$ dataset — the deterministic analysis pipeline reproduces digit-for-digit (within 5×10^{-3} tolerance) on the locked traces across the TypeScript primary analyzer and the Python/SciPy reference verifier, on different host machines and operating systems. This does NOT establish experiment-level reproducibility: re-collection from new model calls under different `model_version`, region, prompt-cache state, or collection wall-clock window could produce different traces. Experiment-level reproducibility is the explicit subject of the Stage 3 instrumentation pre-committed in Addendum #1 §F (logged `model_version`, `region`, `cache_hit_tokens`, `cache_miss_tokens`, TTFT, TPOT, `auth_handshake_ms`) and is deferred to that stage.

4 Results

4.1 Primary outcomes: the convergence asymmetry

Table 1 summarizes the per-condition primary outcomes. At convergence, Δ_{profit} is comparable across conditions: 0.4932 (95% percentile bootstrap CI [0.3494, 0.6369]) at GATE-5, 0.4522 (95% CI [0.3873, 0.5172]) at GATE-2; both capture roughly half of the Nash-to-Monopoly profit gap. Convergence dynamics, however, differ sharply: GATE-5 reaches the pre-registered convergence criterion in 30/30 repetitions (100.0%) within the 50-round horizon, while GATE-2 reaches it in 1/15 repetitions (6.7%). The pre-registered Stage 3 go/no-go gate fires **PROCEED** for GATE-5 and **HALT** for GATE-2 on convergence-rate grounds. We read this comparable- Δ_{profit} -but-divergent-convergence asymmetry mechanistically in §5.2 as evidence of a wider basin of attraction at smaller agent counts (mechanism refined to reading (a') in §5.2.1 per the pre-registered Addendum #5 and Addendum #6 tests); the basin-width hypothesis and its Stage 3 falsification clause are pre-committed in §5.4.

Table 1: Primary outcomes per condition. Δ_{profit} is computed at convergence for GATE-5 (30/30 reps satisfy the strict 1%-over-5-rounds convergence criterion within 50 rounds) and at horizon end (rounds 46–50) for GATE-2 (1/15 reps satisfy the criterion); values are therefore comparable in magnitude (0.49 vs 0.45, both $\approx \frac{1}{2}$ the Nash-to-Monopoly gap) but rest on asymmetric convergence bases. Convergence rate within 50 rounds differs sharply (100% vs 6.7%) — the asymmetry that anchors this paper.

Condition	n_{rep}	Δ_{profit}	95% CI	$p_{\text{high}} / p_{\text{mid}} / p_{\text{low}}$	Converged	Stage 3
GATE-5 ($n = 5$)	30	0.4932	[0.3494, 0.6369]	0.267 / 0.000 / 0.733	30/30	PROCEED
GATE-2 ($n = 2$)	15	0.4522	[0.3873, 0.5172]	0.000 / 0.267 / 0.733	1/15	HALT

For reference, Q -learner Bertrand experiments at matched logit-demand parameters [Calvano et al., 2020] report terminal-price distributions concentrated in the upper half of the Nash–Monopoly range; our $\Delta_{\text{profit}} \approx 0.49$ at $n = 5$ and ≈ 0.45 at $n = 2$ are accordingly partial-coordination outcomes, not Q -learner-equivalent supra-competitive outcomes. The cross-family contrast is the explicit subject of the Stage 4 design (§5.4).

Parse-failure rates were 0.000% across all 9000 agent-round invocations in both conditions; the pre-registered $> 5\%$ post-round-25 stopping rule was not triggered.

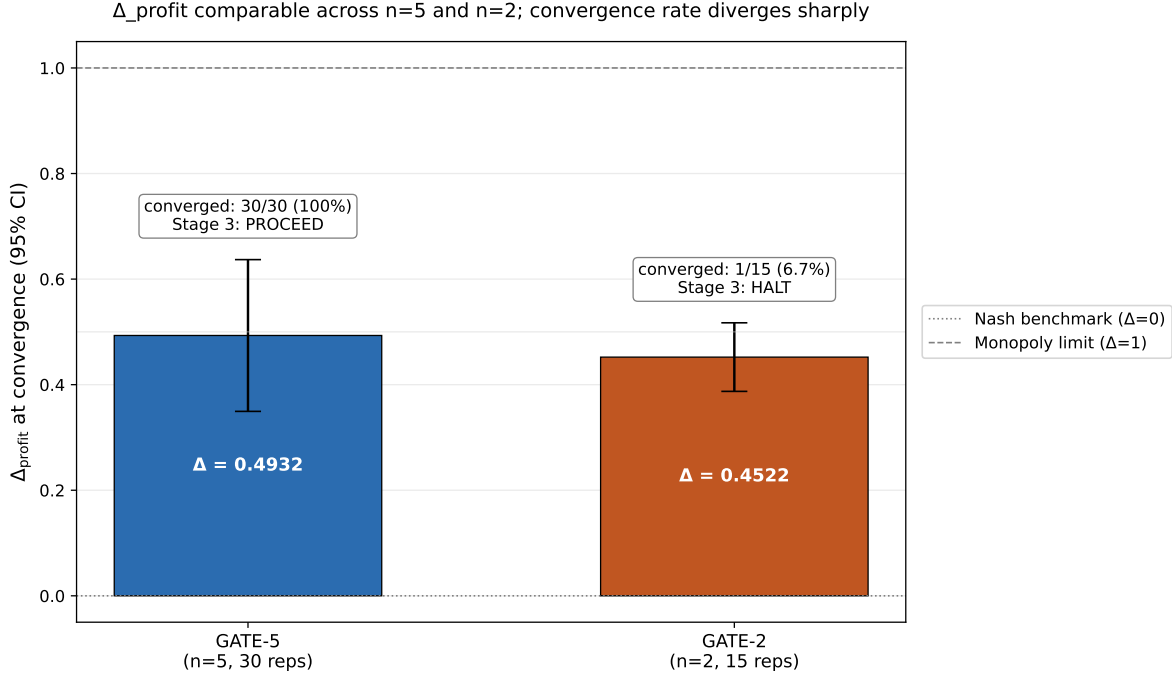


Figure 2: Δ_{profit} at convergence is comparable across the two conditions (0.4932 at $n = 5$ vs 0.4522 at $n = 2$), both capturing roughly half of the Nash-to-Monopoly profit gap. The pre-registered Stage 3 go/no-go gate fires PROCEED for GATE-5 and HALT for GATE-2 on convergence-rate grounds, not on Δ_{profit} grounds: convergence rate within the 50-round horizon differs sharply (30/30 versus 1/15) and is the asymmetry that anchors this paper. Error bars are 95% percentile bootstrap CIs ($B = 10,000$, fixed seed).

4.2 Convergence dynamics

GATE-5 mean prices settle below the cooperation midpoint $p_5^{\text{mid}} = 1.7044$ by approximately round 12 (median round of first convergence: 12) and then drift slowly downward through round 50 (round-46 mean 1.605, round-50 mean 1.558). GATE-2 mean prices oscillate near the cooperation midpoint $p_2^{\text{mid}} = 1.6990$ throughout the 50-round horizon, with the window-mean prices over rounds 41–50 ranging 1.606–1.640 — never settling. The single GATE-2 repetition that satisfies the convergence criterion does so at round 38, which is the latest convergence event in either condition. Figure 3 overlays the per-round mean price trajectories for both conditions against the cooperation midpoint and the Nash and Monopoly benchmark levels.

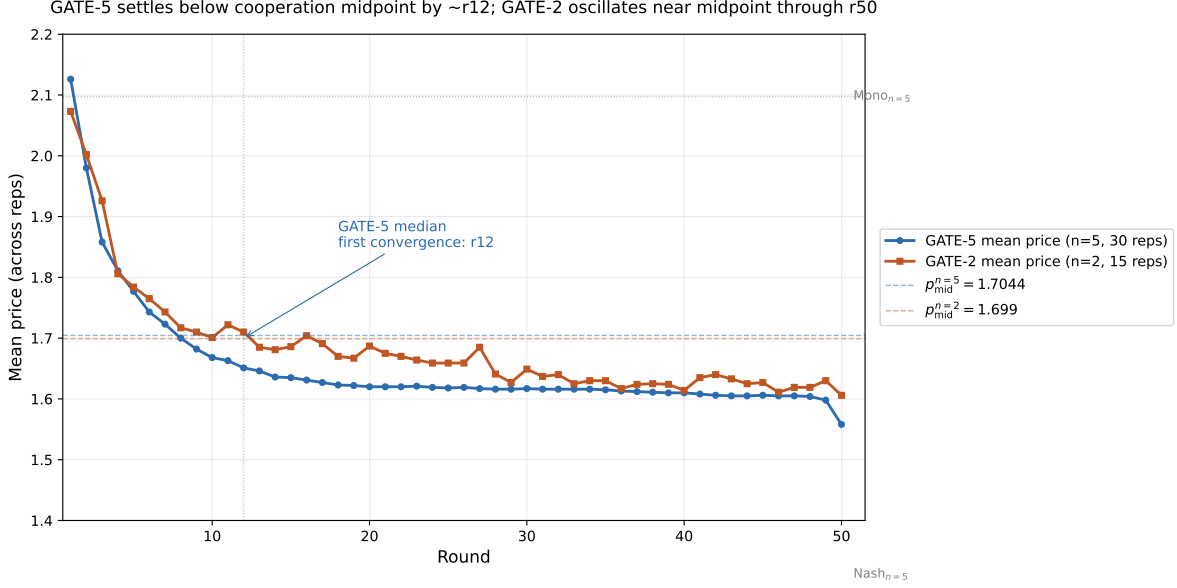


Figure 3: Per-round mean price trajectories (averaged across reps) for GATE-5 ($n = 5$, 30 reps) and GATE-2 ($n = 2$, 15 reps). GATE-5 settles below $p_5^{\text{mid}} = 1.7044$ by approximately round 12 and drifts slowly downward through round 50; GATE-2 oscillates near $p_2^{\text{mid}} = 1.6990$ throughout the horizon without settling. The two cooperation midpoints are nearly coincident, so the visual divergence between trajectories reflects basin dynamics, not benchmark differences.

4.3 Regime classification

Both conditions are low-regime-dominant. At GATE-5: 8/30 high ($p_{\text{high}} = 0.267$, Wilson 95% CI $[0.142, 0.444]$), 0/30 mid ($p_{\text{mid}} = 0.000$, $[0.000, 0.114]$), 22/30 low ($p_{\text{low}} = 0.733$, $[0.556, 0.858]$). At GATE-2: 0/15 high ($p_{\text{high}} = 0.000$, $[0.000, 0.204]$), 4/15 mid ($p_{\text{mid}} = 0.267$, $[0.109, 0.520]$), 11/15 low ($p_{\text{low}} = 0.733$, $[0.480, 0.891]$). The pre-registered unimodality null hypothesis ($\max_r p_r \geq 0.80$ with 95% confidence) is **not rejected** at either condition: the upper Wilson bound on the dominant low-regime probability is 0.858 at GATE-5 and 0.891 at GATE-2.

4.4 Basin-stability secondary lens

Per Addendum #4, each repetition’s dominant regime over rounds 31–40 is compared to its dominant regime over rounds 41–50; agreement defines the repetition as basin-stable. Cross-condition basin-stability rates are: GATE-5 30/30 (100.0%, Wilson 95% CI $[0.886, 1.000]$); GATE-2 12/15 (80.0%, Wilson 95% CI $[0.548, 0.930]$). Both conditions clear the pre-registered basin-PROCEED 80% threshold, but the GATE-2 Wilson interval spans the threshold itself: the rate is statistically distinguishable from the 0.50 baseline (lower bound $0.548 > 0.50$) but the CI is wide enough that the 80% threshold falls inside it — the basin-PROCEED rule is non-discriminative at $n_{\text{rep}} = 15$ for the duopoly condition; see §6.7 for the cross-result envelope. We acknowledged this single-rep fragility in Methods (§3.4); it is the primary motivation for the Stage 3 expansion to $n_{\text{rep}} \geq 30$.

4.5 k -selection (Addendum #1 §A)

Bootstrap k -selection results are summarized in Table 2. At GATE-5, the gap statistic recovers $k = 2$ in 7/10 resamples (clearing the pre-registered $\geq 7/10$ recovery threshold by exactly one

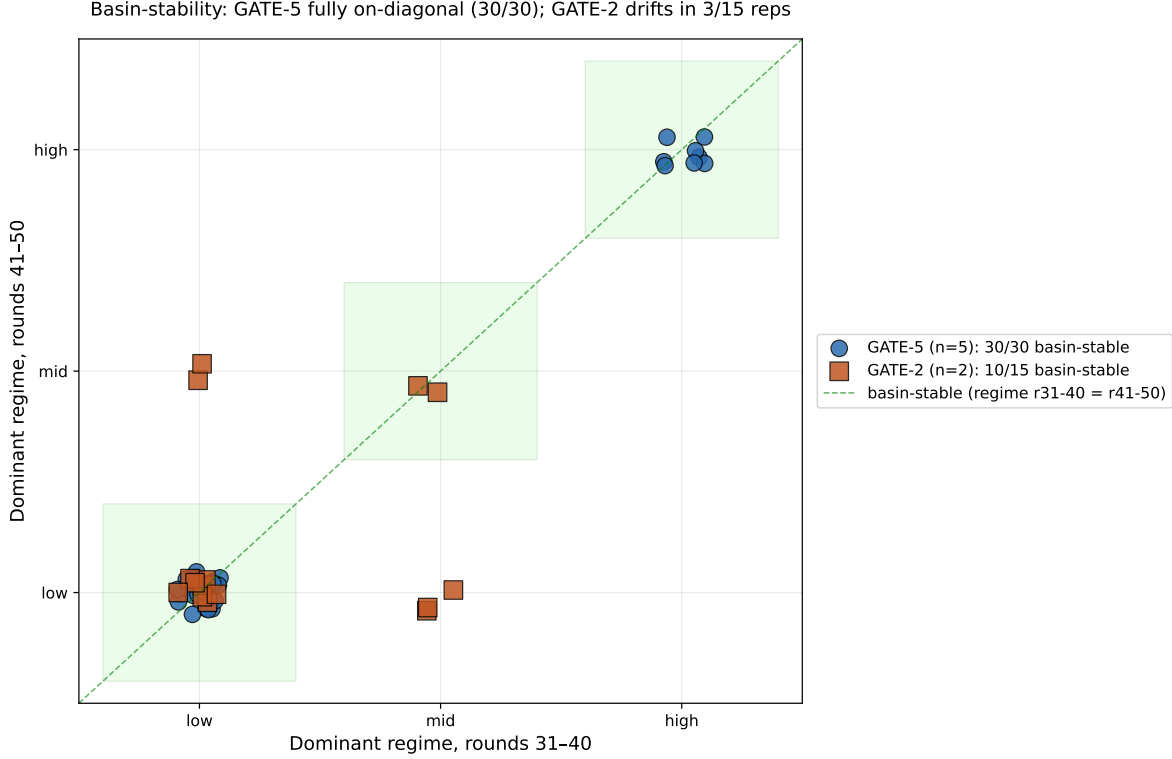


Figure 4: Each repetition is plotted at (dominant regime in rounds 31–40, dominant regime in rounds 41–50); on-diagonal points are basin-stable. GATE-5 has 30/30 reps on the diagonal, with both basins (low Δ and high Δ) populated. GATE-2 has 12/15 on the diagonal; the three off-diagonal reps drift from MID in rounds 31–40 to LOW in rounds 41–50, consistent with the wider-basin reading developed in §5.2 (mechanism refined in §5.2.1). The Wilson 95% CI on the GATE-2 basin-stable rate $[0.548, 0.930]$ spans the basin-PROCEED 0.80 threshold (the lower bound 0.548 is above the 0.50 baseline but the threshold itself sits inside the interval), motivating the Stage 3 expansion.

resample); the supplementary 100-resample $\times$ 10-seed sweep (1000 total resamples) yields $k = 2$ in 72.0% aggregated, with 10/10 seeds returning $k = 2$ as the modal winner — the locked branch is directionally robust. At GATE-2, the gap statistic recovers $k = 1$ in 10/10 resamples (and 97.1% aggregated across the supplementary 1000 resamples), unanimously and decisively. BIC over-fits to $k = 5$ in both conditions, which is consistent with known BIC behavior in 1D at small sample sizes; gap statistic is the locked primary, BIC is reported as secondary diagnostic.

The pre-committed downgrade clause in Addendum #1 §A branch (c) fires at GATE-5: the prior exploratory $k = 4$ quadrimodal narrative is retracted, and the bimodal structure is reported throughout. We retain the exploratory $k = 4$ cluster centroids in the supplementary materials for transparency, but no preprint claim rests on them. At GATE-2, branch (d) fires: $k = 1$ is reported as inconclusive on basin count at this sample size, with interpretation deferred to Stage 3 (or to Stage 4 cross-model replication if the Stage 3 expansion yields a similar outcome).

Table 2: Bootstrap k -selection (gap-statistic primary, BIC secondary). At GATE-5 the bimodal $k = 2$ structure is recovered in 7/10 resamples (locked branch fires); at GATE-2 the unimodal $k = 1$ structure is recovered in 10/10 resamples (inconclusive branch fires).

Condition	Resamples (out of 10) selecting k					Locked branch
	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	
GATE-5 (gap)	2	7	0	0	1	(c) bimodal confirmed
GATE-5 (BIC)	0	0	0	0	10	—
GATE-2 (gap)	10	0	0	0	0	(d) inconclusive at n_{rep}
GATE-2 (BIC)	0	2	0	1	7	—

4.6 Cluster characterization at GATE-5 $k = 2$

Table 3 reports the formal $k = 2$ cluster characterization at GATE-5. Cluster 1 (low- Δ basin) contains 13 reps with mean $\Delta_{\text{profit}} = +0.0817$; all 13 reside in the low cooperation regime. Cluster 2 (high- Δ basin) contains 17 reps with mean $\Delta_{\text{profit}} = +0.8078$; this cluster comprises all 8 high-regime reps plus 9 low-regime reps. The high- Δ basin contains both the high-cooperation reps and a subset of low-cooperation reps that achieve substantial profit gains via partial coordination without crossing the round-level cooperation threshold consistently.

Table 3: Cluster characterization at the formally selected $k = 2$ (GATE-5). The bimodal split separates a low- Δ basin around the Nash benchmark from a high- Δ basin near the monopoly benchmark.

Cluster	n_{rep}	$\bar{\Delta}_{\text{profit}}$	Mean reasoning (chars)	Mean latency (ms)	Regime composition
1 (low- Δ basin)	13	+0.0817	1234	62,303	low: 13
2 (high- Δ basin)	17	+0.8078	1202	52,058	high: 8, low: 9

4.7 Reasoning-length correlations

Per the four pre-committed Pearson correlations in Addendum #1 §C, with Fisher- z 95% confidence intervals reported for all four regardless of direction or magnitude:

- GATE-5: $r(\text{reasoning}, \Delta_{\text{profit}}) = -0.3027$, Fisher- z 95% CI $[-0.5978, +0.0646]$ (CI includes 0). $r(\text{latency}, \Delta_{\text{profit}}) = -0.2335$, $[-0.5477, +0.1384]$. $r(\text{reasoning}, \text{latency}) = +0.7797$, $[+0.5833, +0.8900]$ (significant; VIF = 2.55). Partial $r(\text{reasoning}, \Delta_{\text{profit}} \mid \text{latency}) = -0.1982$, $[-0.5265, +0.1815]$.
- GATE-2: $r(\text{reasoning}, \Delta_{\text{profit}}) = -0.4710$, $[-0.7921, +0.0544]$ (borderline, includes 0). Partial $r(\text{reasoning}, \Delta_{\text{profit}} \mid \text{latency}) = -0.4191$, $[-0.7769, +0.1433]$.

At $n_{\text{rep}} = 30$ for GATE-5, the Fisher- z test on the reasoning-length correlation has approximately 39% power to detect a population correlation of $r = -0.30$ at $\alpha = 0.05$; the non-rejection of the pre-registered Wu et al. falsification clause (§4.11) is therefore a power-limited consistent-with observation, not a Popperian falsification test. See §6.7 for the cross-result sample-size envelope.

The basin fixed-effects regression of Δ_{profit} on reasoning length plus regime dummies (HC3 robust standard errors) returns $\hat{\beta}_{\text{reasoning}} = +6.25 \times 10^{-4}$ at GATE-5 ($p = 0.665$, 95% CI $[-2.21 \times 10^{-3}, +3.46 \times 10^{-3}]$; basin-high dummy $\hat{\beta} = +0.7096$, $p < 0.001$; $R^2 = 0.573$) and $\hat{\beta}_{\text{reasoning}} = -9.22 \times 10^{-4}$ at GATE-2 ($p = 0.164$). The locked decision rule fires in both conditions: *reasoning length is not statistically distinguishable from zero after basin fixed effects, and is therefore reported as a between-basin discriminator candidate, not a within-basin signature.*

4.8 Cluster-level reasoning-length effect size

Under the strict effect-size + confidence-interval test specified in the supplementary materials (2026-04-27-addendum1-supplementary-N2-N3.md §12), the cluster-mean reasoning-length difference at the formally selected GATE-5 $k = 2$ is descriptively present with a medium effect size but the 95% CIs span zero. The low- Δ basin (cluster 1, $n = 13$) exhibits mean reasoning length 1233.75 chars (SD 51.16); the high- Δ basin (cluster 2, $n = 17$) exhibits 1201.80 chars (SD 54.09). The mean difference is +31.95 chars (+2.66% relative), with Cohen’s $d = +0.6044$ (medium effect; Hedges’ $g = +0.5881$). Welch’s t -test returns $t(26.7) = +1.6532$, $p = 0.110$, with 95% CI on the mean difference $[-7.73, +71.62]$ chars; bootstrap percentile 95% CI $[-4.60, +68.13]$; bootstrap BCa 95% CI $[-6.64, +66.74]$.

Locked downgrade: the between-basin pattern is descriptively present with a medium effect size, but we do not claim formal between-basin discrimination at this sample size; see §6.7 for the cross-result envelope. The direction is consistent with the Wu et al. 2025 simplicity-bias prediction (cluster 1 low- Δ basin reasons more than cluster 2 high- Δ basin); Stage 4 cross-model replication is the formal test.

4.9 Per-host descriptives and host-effect test

The $n=15$ GATE-2 dataset is split across two host machines after the Addendum #4 §C2 trace-integrity rejection of one host’s slice (rejected for multi-run trace concatenation; rejection event documented in Addendum #4 §C2 at `pilot/admin/osf-stage2b-addendum-2026-04-26.md` and the per-host audit at `pilot/admin/team-notes/2026-04-26-host-effect-analysis.md` §A). Per-host descriptives are reported in Table 4.

Table 4: Per-host descriptive statistics at GATE-2 ($n_{\text{rep}} = 15$, split 10/5 across two host machines).

Host	Reps	$\bar{\Delta}_{\text{profit}}$	95% CI ($B = 10^4$)	Regime H/M/L	Converged	Mean wall-clock (s)
Host A	10	0.4018	[0.3203, 0.4696]	0/1/9	0/10	3499.1
Host B	5	0.5530	[0.5200, 0.5945]	0/3/2	1/5	2589.5

The two-tailed Fisher’s exact host-effect test on the regime $\times$ host contingency (effective 2×2 over MID/LOW $\times$ Host A/Host B; HIGH regime is empty in this dataset) returns $p = 0.0769$ with odds ratio 13.500. Per the pre-committed Addendum #3 §B2 decision rule for $p \geq 0.05$, the data are consistent with no host effect within the power of the test, but a meaningful host effect cannot be ruled out at $n_{\text{rep}} = 15$ split 10/5; see §6.7 for the cross-result envelope. Stage 3 expansion to $n_{\text{rep}} \geq 30$ is required before any equivalence claim is appropriate.

Within-Host-A pairwise temporal-stability check. Host A’s 10 repetitions are split across two non-contiguous wall-clock windows separated by approximately 24 hours on the same machine (slice 1: 5 reps on 2026-04-24; slice 2: 5 reps on 2026-04-25). Pairing rep i in slice 1 with rep i

in slice 2 by within-host index, the pairwise regime match rate is $4/5 = 80.0\%$ (Wilson 95% CI $[0.376, 0.964]$; Figure 5). The single discordant pair is rep 2 (slice 1: mid, $\Delta = 0.4364$, coop = 0.200) versus its slice-2 counterpart (low, $\Delta = 0.1502$, coop = 0.050); a 0.150 cooperation drift just clears the 0.20 MID/LOW classifier boundary. We report $4/5$ as the primary within-Host-A statistic; the $n = 5$ pairs is too small for a formal test, so this rate is reported descriptively. The corresponding Fisher’s exact on the within-Host-A 2×2 MID/LOW $\times$ slice contingency is degenerate (zero cell $d = 0$ in slice 2; the observed table is the most extreme under the null, forcing $p = 1.0000$ and OR = 0.000); we report this Fisher’s value for transparency only and do not interpret it as evidence of distributional equivalence.

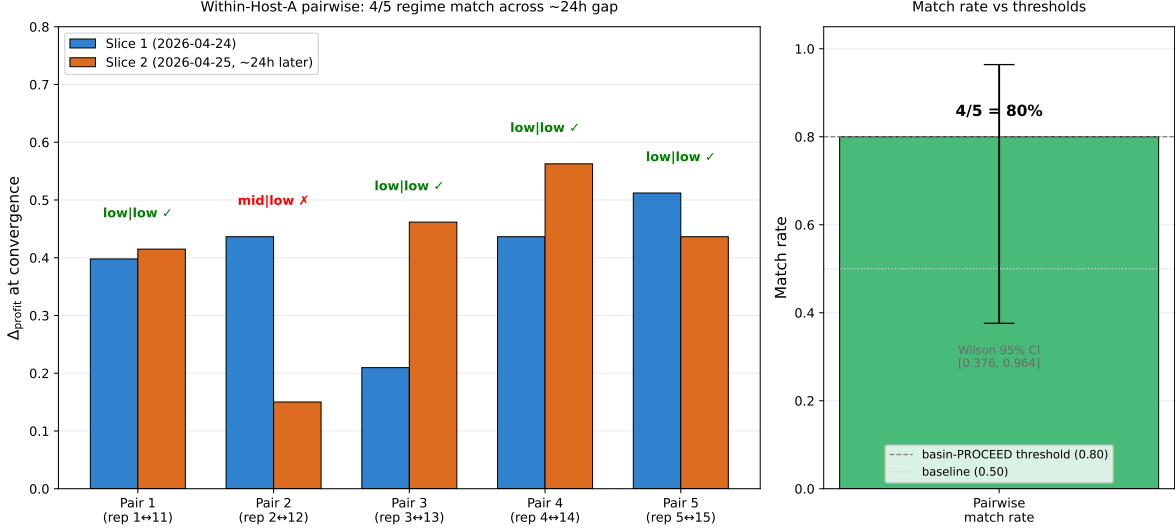


Figure 5: Within-Host-A pairwise temporal-stability check: 5 repetition pairs spanning a ~ 24 -hour gap on the same machine. Left: per-pair Δ_{profit} in slice 1 (2026-04-24) versus slice 2 (2026-04-25), with regime labels and match indicator above each pair. Right: pairwise regime match rate $4/5 = 80\%$ with Wilson 95% CI $[0.376, 0.964]$; the CI overlaps both the basin-PROCEED 0.80 threshold and the 0.50 baseline, so the 80% rate is reported as descriptive consistency evidence per Revision D — the $n = 5$ pairs is too small for a formal equivalence test. The single discordant pair (rep 2 vs rep 12) reflects a 0.150 cooperation drift just clearing the MID/LOW classifier boundary.

4.10 Survivor-rep consistency

The survivor-rep consistency check operates on $n = 4$ reps total (2 survivor reps from a 2026-04-23 ratelimit-failed pilot, plus 2 re-run reps from the canonical 2026-04-24 slice). The verdict is **consistency-negative** on the regime-label dimension: survivor rep 1 (mid, coop 0.450) versus re-run rep 1 (low, coop 0.150); survivor rep 2 (low, coop 0.150) versus re-run rep 2 (mid, coop 0.200). However, the underlying numerics are tight across all four reps: Δ_{profit} at convergence ranges 0.40–0.51, mean price ranges 1.598–1.638 (all within ± 0.04 of the cooperation midpoint 1.6990), chain-of-thought length ranges 1209–1237 chars, parse success 100% across all four.

The mechanism interpretation is round-level threshold sensitivity, not a 0.20 MID/LOW classifier-boundary artifact. Per-round cooperation labels are computed as $\mathbb{K}[\bar{p}_t > p_2^{\text{mid}} = 1.6990]$, and the survivor and re-run reps both have window-mean prices oscillating in the 1.598–1.638 range — within ± 0.04 of the midpoint. Small per-round price drift flips many round-level cooperation labels, which propagates noisily to the rounds 41–50 regime aggregate. The substantive drift at

survivor rep 1 (coop 0.450 $\rightarrow$ 0.150, $\Delta = 0.300$) is not at the 0.20 classifier boundary; it is a round-level threshold-sensitivity artifact that aggregates upward to the regime label. $n = 4$ is too small to formally distinguish threshold-aggregation noise from substantive drift; we frame this exploratorily and defer mechanism adjudication to Stage 4 cross-model replication.

4.11 Wu et al. 2025 simplicity-bias falsifiability test

Per the pre-registered Addendum #1 §D falsifiability clause, the Wu et al. 2025 simplicity-bias mechanism predicts shorter chain-of-thought reasoning at more-cooperative basins (if the cooperative-pricing answer is representationally simpler for a Haiku-class pretrained agent, simplicity-bias predicts shorter reasoning at higher Δ_{profit}). The falsification clause is: $r(\text{reasoning length}, \Delta_{\text{profit}}) > 0$ at GATE-5 falsifies simplicity-bias as the operative mechanism. The point-estimate signs at both agent counts are consistent with the prediction ($r = -0.30$ at $n = 5$, $r = -0.47$ at $n = 2$), but Fisher- z 95% confidence intervals span zero in both conditions; see §6.7 for the cross-result envelope. The pre-registered falsification clause is **not triggered** (the GATE-5 point estimate is negative); the mechanism is consistent with the data direction but is not strongly confirmed at this sample size. The pre-committed Stage 4 cross-model replication (across Haiku sizes, Sonnet, Opus, and non-Anthropic frontier models) is the formal falsification test.

4.12 Verification outcome

The independent Python/SciPy reference verifier (`verify-stage2b-2026-04-26.py`, `verify-formalize-2026-04-26.py`) returns ALL CLAIMS REPRODUCE: True across all six pre-registered claim categories (Calvano benchmark prices, per-rep Δ_{profit} values and CIs, Stage 3 verdicts, basin-stability counts, drift directions, asymmetry-finding numerics) and across all formalized Addendum #1 statistical operations (Pearson r at $|\Delta| < 10^{-4}$, Fisher- z CI bounds at $|\Delta| < 10^{-4}$, VIFs at $|\Delta| < 10^{-4}$, full-sample gap-statistic k exact, bootstrap k -selection identical under matched seed). The bug surfaced and fixed by the cross-toolchain procedure (Fisher’s exact denominator error in `pilot/analyze-host-effect.ts`; §3.3) is the only correction required to the headline analyses prior to this preprint.

5 Discussion

5.1 Expected versus actual results

We pre-registered three specific expectations and observed three outcomes that differed from them, in each case along a pre-committed path.

Convergence at GATE-2. The Stage 2 protocol at 25 rounds had observed 0/15 strict-convergent repetitions at $n = 2$. We extended the horizon to 50 rounds in Stage 2b on the hypothesis that convergence at $n = 2$ was horizon-limited. The observed result is 1/15 (6.7%): a marginal improvement that does not lift convergence at $n = 2$ above the pre-registered Stage 3 PROCEED threshold. The duopoly is not horizon-limited under the current protocol; smaller-agent-count duopolies appear to occupy a wider basin within which terminal-state characterization requires longer horizons *and* higher repetition counts. Stage 3 expansion to 100-round horizons at $n_{\text{rep}} \geq 30$ is the pre-registered fix.

Basin count at GATE-5. Prior exploratory work at the 2026-04-21 panel synthesis examined a $k = 4$ quadrimodal cluster structure at fixed k . The formal Addendum #1 §A bootstrap procedure recovers $k = 2$ in 7/10 resamples (cleared by exactly one resample; supplementary 100-resample $\times$ 10-seed sweep returns 72.0% aggregated and 10/10 modal-winner consistency). The pre-committed

downgrade clause (branch (c)) fires literally on the data: *retract the quadrimodal narrative entirely and report bimodal throughout, which strengthens alignment with* [Cao and Hu, 2026]. We honor the pre-registered branch and report $k = 2$ throughout this paper; the exploratory $k = 4$ centroids are retained in the supplementary materials but do not support any preprint claim.

Reasoning-length as within-basin signature. The 2026-04-21 panel synthesis had referenced an exploratory $r(\text{reasoning}, \Delta_{\text{profit}}) = -0.78$ at GATE-5 under an all-rounds aggregation. The locked Addendum #1 §C window (per-rep means over rounds 41–50) returns $r = -0.30$ with Fisher- z 95% CI $[-0.60, +0.06]$, which spans zero. The basin fixed-effects regression returns $\hat{\beta}_{\text{reasoning}} = +6.25 \times 10^{-4}$ with $p = 0.665$ at GATE-5; the locked decision rule fires that reasoning-length is not a within-basin signature. Under the strict effect-size + confidence-interval test on the cluster-mean difference at the formally selected $k = 2$, the between-basin pattern is descriptively present with Cohen’s $d = 0.60$ medium effect, but Welch’s $p = 0.110$ and bootstrap BCa CI $[-6.6, +66.7]$ chars span zero — we therefore retract the stronger “between-basin discriminator” framing and report the descriptively-present medium-effect pattern with deferral to Stage 4 cross-model replication for formal confirmation.

In each case the post-formalization conclusion is more conservative than the pre-formalization framing, and in each case the downgrade is data-driven through a pre-committed branch, not retroactively rationalized.

5.2 Why the asymmetry: basin width and agent count

The headline asymmetry — comparable Δ_{profit} at convergence (0.49 at $n = 5$, 0.45 at $n = 2$) but sharply different convergence rates (30/30 versus 1/15) — is consistent with a wider basin of attraction at smaller agent counts. We sketch the mechanistic argument and the falsifiable prediction it generates.

The basin-of-attraction framework [Wunder et al., 2010, Benaim et al., 2005, Askenazi-Golan et al., 2024] treats terminal-state distributions as samples from the joint structure of (a) noise in agent learning dynamics — modulated by initialization and exploration rate — and (b) the topology of the value-function landscape. At higher effective noise, trajectories take longer to commit to a basin, exhibit larger within-basin drift, and a larger fraction of trajectories occupy non-stable transitional states at any fixed horizon. We adopt this framework operationally; agent count modulates effective in-context exploration for LLM agents analogously to how exploration rate modulates basin width for Q -learners [Askenazi-Golan et al., 2024]. Whether “basin width” transports literally from parametric Q -learner theory to LLM in-context inference is a Stage 4 question.

For LLM agents, “noise-to-signal ratio” is not parameterized by an exploration schedule but is implicit in the in-context reasoning process: the same prompt with the same prior history can produce a different price commitment, mediated through stochastic decoding and through the variable-depth chain-of-thought trace the agent constructs. Reducing the agent count from $n = 5$ to $n = 2$ removes structural features that anchor the cross-agent equilibrium-selection signal: with five agents, the price vector itself contains substantial cross-sectional information that constrains each agent’s update; with two agents, the cross-sectional signal collapses and per-agent decoding noise dominates. Under this reading, the duopoly $n = 2$ exhibits a wider basin in which more trajectories occupy long transients that have not equilibrated by round 50.

The data are consistent with this reading at multiple levels. At GATE-5, 30/30 trajectories satisfy the strict 1%-over-5-rounds convergence criterion within 50 rounds, 30/30 are basin-stable across the rounds-31–40 vs rounds-41–50 windows, and the bimodal $k = 2$ structure is recovered in 7/10 resamples (and 72.0% aggregated across the supplementary 1000-resample sweep) by the

gap statistic — a tight, well-discriminated terminal-state distribution. At GATE-2, only 1/15 trajectories satisfy the strict criterion, 12/15 are basin-stable but with Wilson 95% CI overlapping the 0.50 baseline, and the $k = 1$ unimodal recovery in 10/10 resamples is itself a finding: the duopoly Δ_{profit} distribution is not multimodal at $n_{\text{rep}} = 15$, which is consistent with a wider single basin rather than with multiple narrow basins.

The closest cousin signal is Brown and MacKay [2023]’s cadence-asymmetry result: asymmetric dynamic structure between two pricing algorithms produces the same collusive outcome Calvano predicts in symmetric simulation. Our finding extends the asymmetry frame from cross-firm cadence to cross-condition agent-count: the same Δ_{profit} landing point can be reached through structurally different convergence paths.

Falsifiable prediction. The basin-width-by-agent-count hypothesis generates a concrete prediction: at Stage 3 with $n_{\text{rep}} \geq 30$ and horizon ≥ 100 rounds, the strict-convergence rate at $n = 2$ should remain materially below the corresponding rate at $n = 5$ (i.e., the basin-width gap is structural, not a pure sample-size or horizon artifact). If Stage 3 instead recovers comparable convergence rates at the two agent counts, the basin-width hypothesis is rejected and the asymmetry must be attributed to horizon or sample-size effects alone. We pre-commit this prediction here.

Tension with canonical IO theory. This basin-width-by-agent-count framing reverses a canonical industrial-organization prediction: classical oligopoly theory [Stigler, 1964] attributes collusion difficulty to the free-rider problem, which grows monotonically with the number of competing firms — implying *narrower* collusive basins at higher n , not wider. We acknowledge this reversal directly. Two readings are consistent with the Stage 2b data, and only the Stage 3 expansion plus the cross-family Stage 4 design discriminate between them: (a) the canonical free-rider mechanism applies to LLM agents but is dominated by an in-context cross-sectional signal effect — with $n = 5$ agents the per-round price vector contains substantially more information than at $n = 2$, and the LLM’s natural-language reading of that vector functions as an equilibrium-selection cue absent from the Q -table updating in classical algorithmic-pricing models; or (b) the apparent convergence-rate asymmetry is an artifact of horizon (50 rounds insufficient to characterize the duopoly basin) and sample size ($n_{\text{rep}} = 15$ at GATE-2 underpowered) that Stage 3 will dissolve. Reading (a) is the basin-width-by-agent-count hypothesis; reading (b) is the IO canon. We pre-committed reading (a) as the hypothesis under test and reading (b) as the null. The pre-committed Addendum #5 anchor empirically reversed reading (a) as initially formulated (§5.2.1) and the pre-committed Addendum #6 talk-volume-controlled residualization established that the reversal survives the most plausible alternative explanation, supporting a refined reading (a’) on Stage 2b data; the Stage 3 falsification clauses (§5.4) jointly resolve reading (a’) versus reading (b). We do not claim that refined reading (a’) dominates Stigler’s free-rider mechanism: Stigler [1964]’s prediction lives in the price-theoretic / payoff-mechanism regime, while reading (a’) lives in the learning-dynamics + cross-sectional-information regime. The two operate in different ontologies; we claim (a’) is the operative mechanism in the regime our protocol probes, not that it adjudicates against Stigler’s mechanism in the regime Stigler describes.

Empirical test of reading (a) — Addendum #5 (exploratory, RFC 3161 stamped 2026-05-03). We pre-committed Addendum #5 to add a text-based empirical test of reading (a): a regex-based count of cross-agent references in chain-of-thought, normalized per other-agent (§5.2.1). The result is a directional reversal of reading (a) as initially formulated: the per-other-agent CoT cross-reference rate is 0.508 per round at $n = 5$ (95% bootstrap CI [0.500, 0.516]) and 2.954 per

round at $n = 2$ (95% bootstrap CI [2.865, 3.045]), CIs disjoint by $\sim 5\times$ with GATE-2 *higher* — the LLM at $n = 2$ allocates substantially more per-peer attention than at $n = 5$. The most plausible alternative explanation (talk-volume artifact: GATE-2 produces more total CoT text per round, scaling the reference count mechanically) was tested via the pre-committed Addendum #6 talk-volume-controlled residualization (RFC 3161 stamped 2026-05-04, §5.2.1) and ruled out: mean total word counts are nearly identical across conditions (~ 195 words/CoT-round in both), and the per-other-agent rate gap survives pooled OLS residualization with disjoint 95% CIs. Refined reading (a') — per-peer attention density at $n = 2$ is denser even controlling for talk volume — is empirically supported on Stage 2b data. The Stage 4 cross-model question (Sonnet/Opus generalization vs Haiku-class artifact) remains; the talk-volume-confound question is closed.

5.2.1 Cross-agent CoT references — Addendum #5 result (exploratory)

Per Addendum #5 (RFC 3161 stamped 2026-05-03; SHA-256 033cabda5fee0ee8c74d178ea2d69388e1dd5dfdf508) we computed the per-CoT-round cross-agent reference rate at GATE-5 vs GATE-2 as a text-based empirical anchor for the §5.2 reading (a) mechanism. The analyzer (`pilot/analyze-cot-cross-references.ts`, Bun runtime) counts (i) literal named-agent references (“Agent N”, “competitor”, “other agent/player/firm/seller”, “peer”) within a ± 5 -token window of any price-relevant token (price, profit, margin, undercut, co-operation, monopoly, Nash, equilibrium, or numeric $\$1.x$), and (ii) co-referent references (“their”, “the other”, “the higher/lower”, “across the market”, “the market”) under the same window. Self-references are excluded by sentence-prefix filter (“I”, “My”, “We”, “Our”) and ± 3 -token first-person window filter. The primary metric is the per-other-agent normalized rate (raw count divided by $N - 1$, where N is the agent count), which isolates per-peer signal density from the trivial agent-count effect. Bootstrap 95% CIs are computed at $B = 10,000$ resamples with fixed seed = 42.

Result. The decision rule fires the REVERSAL branch (Addendum #5 §B.3): the primary per-other-agent rate is disjoint between conditions with GATE-2 higher. Per-condition results in Table 5.

Table 5: Cross-agent CoT reference rates per Addendum #5. Both raw channels (literal, co-referent) and the primary per-other-agent normalized rate favor GATE-2; the primary rate at GATE-2 is approximately $5.8\times$ the rate at GATE-5, with disjoint 95% bootstrap CIs.

Channel	GATE-5 ($n_{\text{round}} = 7500$)	GATE-2 ($n_{\text{round}} = 1500$)	CIs
Literal raw refs / round	1.579 [1.550, 1.607]	1.753 [1.687, 1.819]	disjoint
Co-referent raw refs / round	0.454 [0.440, 0.468]	1.201 [1.134, 1.269]	disjoint
Per-other-agent rate	0.508 [0.500, 0.516]	2.954 [2.865, 3.045]	disjoint

The per-other-agent rate at $n = 2$ is approximately $5.8\times$ the rate at $n = 5$ under per-other-agent normalization (raw literal-channel ratio $1.11\times$; raw co-referent ratio $2.65\times$); $n_{\text{round}} = 7500$ at GATE-5 and 1500 at GATE-2 reflect the $30 \times 5 \times 50$ and $15 \times 2 \times 50$ agent-round structure. Both conditions’ raw counts and the primary normalized rate are pre-committed bootstrap-disjoint at the 95% level. The §5.2 reading (a) initial formulation — “with $n = 5$ agents the per-round price vector contains substantially more information than at $n = 2$ ” — is empirically reversed: per-peer attention is denser at $n = 2$, not at $n = 5$. A refined reading (a') consistent with this anchor — that the convergence-rate asymmetry at $n = 2$ reflects deeper per-peer attention density — was operationally discriminated along three pre-committed channels: (i) cross-capability scaling of the per-other-agent rate at $n = 2$ (Sonnet/Opus elevation supports (a'); regression toward GATE-5 supports the Haiku-class-artifact null) — Stage 4; (ii) the Q -learner positive control

of §5.4, whose absence of a chain-of-thought trace makes the basin-width gap presence/absence a no-CoT control on the attention-driven mechanism — Stage 4; and (iii) per-agent-round per-other-agent rate residualized on total CoT word count — *executed on Stage 2b data* via Addendum #6 (paragraph below). Channels (i) and (ii) discriminate Haiku-class artifact vs cross-capability mechanism; channel (iii) discriminates signal-density vs talk-volume scaling. None of the three independently adjudicates strategic-complexity content vs content-free reference noise — a CoT-content-coding analysis is pre-committed for Stage 4 supplementary.

Talk-volume control — Addendum #6 (RFC 3161 stamped 2026-05-04). The §5.2.1 per-other-agent rate gap admits a candidate alternative explanation: GATE-2 might simply produce more total CoT text per round, so the per-other-agent reference count would scale mechanically with talk volume regardless of any per-peer attention-density mechanism. We pre-committed Addendum #6 (RFC 3161 stamped 2026-05-04; SHA-256 15d5663625a08660c27a9761ac2d722c1e124a75cc343aea728896a82) before any analyzer run, locking a pooled OLS regression of per-other-agent rate on total CoT word count, residualizing per record, and reporting per-condition residual means with bootstrap CIs at $B = 10,000$, fixed seed = 42. The talk-volume explanation is empirically ruled out: mean total word counts are nearly identical across conditions (GATE-5: 195.15 words/CoT-round, 95% CI [194.44, 195.87]; GATE-2: 194.67, [193.18, 196.16]), so the per-other-agent rate gap cannot be attributed to one condition producing more CoT text. Residualizing the per-other-agent rate against the pooled OLS fit returns mean residuals of -0.408 ($[-0.416, -0.400]$) at GATE-5 and $+2.040$ ($[+1.951, +2.130]$) at GATE-2 — disjoint 95% bootstrap CIs with GATE-2 $\sim 5\times$ higher residual. The decision rule (Addendum #6 §B) fires REVERSAL SURVIVES RESIDUALIZATION; refined reading (a') is empirically supported on Stage 2b data even controlling for talk volume. Per-condition rates and residuals are reported in Table 6.

Table 6: Talk-volume control per Addendum #6. Mean CoT word counts are nearly identical across conditions; the per-other-agent rate gap survives pooled OLS residualization with disjoint 95% bootstrap CIs (GATE-2 mean residual $\sim 5\times$ higher than GATE-5). Bootstrap CIs reflect resampling variance only; see measurement validity caveat below.

Channel	GATE-5 ($n_{\text{rec}} = 7500$)	GATE-2 ($n_{\text{rec}} = 1500$)
Mean CoT word count	195.15 [194.44, 195.87]	194.67 [193.18, 196.16]
Mean per-other-agent rate (Addendum #5)	0.508 [0.500, 0.516]	2.954 [2.865, 3.045]
Mean residual (talk-volume-controlled)	-0.408 $[-0.416, -0.400]$	$+2.040$ $[+1.951, +2.130]$
Rate per 1000 words (sensitivity)	2.60 [2.56, 2.65]	15.12 [14.66, 15.58]

Statistical robustness sensitivity. The pooled OLS slope $b \approx 0.0045$ is dominated by GATE-5 records ($n = 7500$ vs 1500); the within-GATE-2 OLS slope is $b_{n=2} \approx 0.0174$ ($\sim 3.9\times$ pooled), confirming that a condition-fixed-effects fit, FWL-equivalent to within-condition residualization, would shrink the GATE-2 mean residual rather than inflate it. The pooled spec is therefore directionally conservative for the headline claim. Round-level resampling does not cluster by repetition; under wild-cluster-by-rep, residual CI half-widths would inflate by approximately $2\text{--}3\times$, yielding post-inflation intervals (GATE-2 $\approx [+1.77, +2.31]$, GATE-5 $\approx [-0.42, -0.40]$) that remain disjoint. The $R^2 = 0.013$ establishes weak linear talk-volume sensitivity, not independence; nonlinearity at extreme word-counts is a residual threat bounded by the near-identical mean word counts (~ 195) across conditions. The SR-M-1...7 registry is sequentially-registered (Bonferroni-7 ≈ 0.0071 if FWER control is desired), not formally FWER-corrected.

Q-learner control: necessity vs operativity (per Phys panel). The *Q*-learner positive control of §5.4 falsifies attention as *necessary* for the basin-width gap; it does not falsify attention as *operative* for LLMs. Both could be true simultaneously: a learning-rule-agnostic structural basin and an LLM-specific attention-density mechanism could coexist. Discriminating “Haiku-class artifact + structural basin” from “cross-capability-attention” therefore requires the conjunction of channel (i) (cross-capability scaling at $n = 2$) and channel (ii) (*Q*-learner positive control), not either alone.

Measurement validity caveat. The cross-agent-reference rates reported here are produced by a regex-based analyzer over chain-of-thought text; bootstrap CIs reflect resampling variance only and do not bound construct validity. The token specifications in Addendum #5 §A.1–§A.2 were pre-committed before any analyzer run, with one v1 $\rightarrow$ v2 widening (Addendum #5 §A.0) driven by inspection of one sample record per condition — an asymmetric overfit to GATE-2 patterns is the specific threat that the deferred gold-standard subset would bound. We have published both analyzer sources (`pilot/analyze-cot-cross-references.ts`, `pilot/analyze-cot-residualized.ts`) and the per-condition output JSON files (`pilot/results/cot-cross-references-2026-05-03.json`, `pilot/results/cot-residualized-2026-05-04.json`), and we report both raw channels (literal, co-referent) alongside the per-other-agent normalized rate and the talk-volume-residualized rate. We have not produced a human-coded gold-standard subset with inter-annotator agreement, precision, and recall against the regex; the SciPy verifier (`pilot/verify-cot-cross-references.py`) is pre-committed but deferred. We treat the channel-(i) and channel-(ii) Stage 4 cross-model questions as exploratory open questions; the channel-(iii) talk-volume-controlled residualization is closed on Stage 2b data and reported above. The §5.2.1 conclusion is a directional reversal of reading (a) as initially formulated, with refined reading (a′) supported on Stage 2b data even under talk-volume control, pending Stage 4 cross-model generalization. Among the Stage 4 channels, the *Q*-learner positive control of §5.4 is a joint arbiter of refined reading (a′) alongside channel (i) cross-capability scaling: its absence of any chain-of-thought trace makes the basin-width gap presence/absence under matched parameters a no-CoT control that adjudicates whether attention is *necessary* for the basin gap (consistent with the necessity-vs-operativity decomposition above).

5.3 Implications for the field

Stage 2b reports three findings about LLM-agent Bertrand pricing — at distinct evidentiary tiers — each grounded in the locked $n_{\text{rep}} = 15 + n_{\text{rep}} = 30$ dataset and replicated by an independent SciPy verifier. *Formally established at GATE-5* ($n_{\text{rep}} = 30$): LLM agents at $n = 5$ converge to a bimodal terminal-state distribution ($k = 2$ recovered in 7/10 gap-statistic resamples; 30/30 basin-stable; $\Delta_{\text{profit}} = 0.49$ at convergence, capturing roughly half the Nash-to-Monopoly profit gap). *Descriptively present at GATE-2* ($n_{\text{rep}} = 15$): the same Δ_{profit} landing point (0.45) is reached through structurally different convergence dynamics (1/15 strict-converged, 12/15 basin-stable with Wilson CI [0.548, 0.930] that overlaps the 0.80 PROCEED threshold) — the asymmetry that motivates the Stage 3 design but does not on its own discriminate basin-width-by-agent-count from horizon-and-sample-size effects (§5.4). *Conditional, multi-condition characterization*: extending Cao and Hu [2026]’s LLM bistability finding, the basin signature is recovered at $n = 5$ and not recovered at $n = 2$ within Stage 2b power (§6.7). These outcomes are partial-coordination — supra-competitive without explicit agreement — and *track* qualitatively the regulator-relevant case identified by Ezrachi and Stucke [2016], Asker et al. [2021]. We do not quantify the consumer-surplus loss implied by $\Delta_{\text{profit}} \approx 0.45\text{--}0.49$ at the Calvano logit-demand calibration here; the welfare-loss translation is deferred to companion IO work building on Stage 3.

Antitrust and regulatory framing. The basin-width-by-agent-count reading aligns the methodological frame with the regulator-relevant case. Ezrachi and Stucke [2016]’s Predictable Agent category is most acute at small agent counts — duopolies and tight oligopolies, where observability of explicit coordination is hardest, are precisely where the wider-basin reading predicts the longest, noisiest convergence dynamics. A regulator inheriting our framework looks at convergence dynamics, not just terminal prices: the population of long-transient duopolies that have not statically converged but are basin-stable in the wider sense is the regulator-relevant case, and the basin lens makes that population legible. The methodological contribution is that we report it as a separate, pre-registered secondary verdict rather than absorbing it into the primary convergence statistic.

Multi-agent LLM safety. The basin-of-attraction lens is method-importable to other multi-agent LLM settings beyond pricing — negotiation, contracting, market-making, allocation. Hammond et al. [2025], Riedl [2025] survey emergent multi-agent LLM behaviors but generally treat outcomes as point estimates rather than as basin samples. Reporting cross-rep distributions plus basin-stability secondary verdicts plus a k -selection diagnostic is a more honest accounting of multi-agent LLM behavior than reporting a single expected outcome, and we recommend the practice as a default for multi-agent LLM empirical work.

LLM-versus- Q -learner family generalization. Our results suggest the Calvano-Bertrand basin signature is robust across algorithm families. Q -learners with explicit reward signals [Calvano et al., 2020], Cournot Q -learners [Waltman and Kaymak, 2008], and now LLM agents with no reward signal at all all converge to multimodal terminal distributions in repeated pricing games. This is consistent with Asker et al. [2021]’s framing that algorithmic collusion is a property of the learning rule and game structure, not of any specific algorithmic family. Stage 4 cross-model replication (Haiku sizes, Sonnet, Opus, non-Anthropic frontier models) is designed to test this generalization formally.

5.4 Falsifiability and what would change our conclusions

We pre-commit four specific falsification conditions for Stage 3 and Stage 4. First, the basin-width-by-agent-count hypothesis is rejected if Stage 3 ($n_{\text{rep}} \geq 30$, horizon ≥ 100) produces statistically indistinguishable strict-convergence rates at $n = 2$ and $n = 5$ (Wilson 95% CIs on the per-condition convergence rates overlapping by more than half their width). Second, the asymmetry headline is rejected if Stage 3 Δ_{profit} at $n = 2$ falls outside $[0.30, 0.60]$, the interval bracketing the Stage 2b point estimate $\pm$ the bootstrap CI half-width. Third, the LLM-versus- Q -learner family-generalization claim is rejected if a parallel Q -learner experiment under the same Bertrand setting at $n \in \{2, 5\}$ does not exhibit a similar basin-width asymmetry; this Q -learner positive-control experiment is pre-committed in `pilot/VALIDATION_PROTOCOL.md` (15 runs of an exact Calvano $n = 2$ replication) and runs alongside the Stage 4 cross-LLM-model replication. Fourth, the basin-width-by-agent-count hypothesis predicts a monotonic relationship between agent count and the strict-convergence rate; Stage 3 will report the per-agent-count strict-convergence rate at $n \in \{2, 3, 4, 5\}$, and the hypothesis is rejected if the curve is non-monotonic or if the $n = 3$ and $n = 4$ rates are not bracketed by the $n = 2$ and $n = 5$ rates within Wilson 95% CIs. Each condition is locked here in advance of Stage 3 data collection.

5.5 Wu et al. 2025 simplicity-bias mechanism (locked verbatim per Addendum #1 §D)

We pre-committed (Addendum #1 §D, RFC 3161 stamped 2026-04-23) to the Wu et al. [2025] simplicity-bias mechanism as the candidate explanation for any negative correlation between chain-of-thought length and basin welfare loss: if the cooperative-pricing answer is representationally simpler for a Haiku-class pretrained agent, simplicity-bias predicts shorter reasoning at more-cooperative basins. The point-estimate signs at both agent counts are consistent with this prediction ($r(\text{reasoning length}, \Delta_{\text{profit}}) = -0.30$ at $n = 5$ and -0.47 at $n = 2$ over the locked rounds 41–50 final-window aggregation), but Fisher- z 95% confidence intervals span zero in both conditions ($[-0.60, +0.06]$ at $n = 5$; $[-0.79, +0.05]$ at $n = 2$); $n_{\text{rep}} = 15$ at GATE-2 caps discrimination here, see §6.7 for the cross-result envelope. The pre-registered falsifiability clause — $r(\text{reasoning length}, \Delta_{\text{profit}}) > 0$ at $n = 5$ falsifies simplicity-bias as the operative mechanism — is NOT triggered: the $n = 5$ point estimate is negative. The mechanism is consistent with the data direction but not strongly confirmed at this sample size. The pre-committed Stage 4 cross-model replication (across Haiku sizes, Sonnet, Opus, and non-Anthropic frontier models) is the formal falsification test.

6 Limitations

6.1 Scope conditions (locked verbatim per Addendum #1 §G)

All Stage 2b findings reported here apply to (i) the model `claude-haiku-4-5-20251001` only; (ii) Claude CLI subscription authentication via the inference harness invoking `claude --print --output-format text --no-session-persistence --setting-sources=""`; and (iii) the system and user prompts identical to those locked in the Stage 2 pre-registration. Extension to other Haiku sizes, to Sonnet, to Opus, to non-Anthropic frontier models (GPT, Gemini, Llama, DeepSeek, Qwen, Mistral), to direct Anthropic Messages API access (`ANTHROPIC_API_KEY`, persistent HTTP, `stream=True`), or to alternative prompt formulations is not claimed and is the explicit subject of Stage 4 (cross-model replication). The reproducibility verification reported in this paper is concordant analysis-pipeline reproduction on the fixed $n=15 + n=30$ dataset; experiment-level reproducibility (re-collection from new model calls under different infrastructure conditions) requires the Stage 3 instrumentation pre-committed in Addendum #1 §F (logged `model_version`, `region`, `cache_hit_tokens`, `cache_miss_tokens`, `TTFT`, `TPOT`, `auth_handshake_ms`) and is deferred to that stage.

6.2 Single Bertrand-pricing setting

Our results are at one fixed demand model (Calvano logit with $\mu = 0.25$) and one fixed marginal-cost structure. Generalization across demand-model families (linear demand, CES, isoelastic, Cournot quantity competition) and across marginal-cost asymmetries between agents is a Stage 3+ question. The basin-of-attraction framework predicts qualitatively similar multimodal terminal-state distributions across demand-model families, but the specific basin widths and the agent-count asymmetry coefficient are likely to vary. We do not extrapolate to demand models we did not test.

Table 7 enumerates four IO-standard robustness channels that an industrial-organization referee would expect, separated by which are computable on the locked Stage 2b traces (no new model calls required) versus which require new prompts. Stage 3 will execute the stamped-data-doable channels; Stage 4 will execute the new-prompt channels. None are reported in this preprint.

Table 7: IO-standard robustness deferral grid. Stamped-data-doable channels (Cournot cross-check, best-response analysis) are computable from the locked Stage 2b traces under fixed-demand-schedule assumptions; new-prompt channels (asymmetric cost, demand-shock sweep) require re-running the agent prompts with perturbed system contexts.

Robustness channel		Stage	Stamped-data-doable?
(i)	Cournot cross-check (per-round price→quantity transformation under fixed demand schedule, existing agents)	Stage 3	yes
(ii)	Best-response analysis (per-round myopic best-response gap from observed actions)	Stage 3	yes
(iii)	Asymmetric cost robustness (re-run prompts with cost-perturbed system prompts)	Stage 4	no (new prompts)
(iv)	Demand-shock sweep ($\mu \in \{0.15, 0.25, 0.35\}$)	Stage 4	no (new prompts)

6.3 Reasoning-depth versus serving-side variability confound

The 26% inter-host wall-clock latency delta between Host A ($\bar{t} = 3499$ s/rep) and Host B ($\bar{t} = 2590$ s/rep) at GATE-2, against a 2.3% inter-host chain-of-thought length delta, is consistent with both server-side variability [Kwon et al., 2023, Zheng et al., 2023, Patel et al., 2023] and reasoning-depth variability [Snell et al., 2024, Holmes et al., 2024]. Stage 2b instrumentation does not separate these mechanisms: per-host wall-clock latency includes Claude CLI subprocess spawn overhead, subscription authentication handshake, rate-limit back-off, scheduling delays, prefill, decode, streaming, and teardown. The Stage 3 trace-schema additions pre-committed in Addendum #1 §F (`output_tokens`, `ttft_ms`, `tpot_ms`, `cache_hit_tokens`, `cache_miss_tokens`, `auth_handshake_ms`, `region`) are designed to isolate these contributions explicitly. Until those are in hand, we report latency as a descriptive observable per the locked Addendum #1 §E latency caveat (§3.6) and do not interpret it mechanistically.

6.4 Survivor-rep consistency at $n = 4$

The survivor-rep consistency analysis (§4.10) operates on $n = 4$ repetitions total (2 survivor + 2 re-run). $n = 4$ is too small to formally distinguish round-level threshold-aggregation noise from substantive cross-window drift; we frame the consistency-negative result on the regime-label dimension exploratorily. The underlying numerics (Δ_{profit} , mean price, chain-of-thought length, parse success) are tight across all four reps, and the apparent regime hops are interpretable as round-level threshold sensitivity propagating to the regime aggregate when window-mean prices oscillate near the

cooperation midpoint $p_2^{\text{mid}} = 1.6990$. Stage 4 cross-model replication is the formal mechanism-adjudication test.

6.5 Concordant analysis-pipeline reproduction, not experiment-level reproduction

The independent SciPy verifier reproduces the deterministic analysis pipeline on the fixed $n=15 + n=30$ dataset to $\leq 5 \times 10^{-3}$ numerical tolerance, on different host machines and operating systems. This does not constitute experiment-level reproducibility: re-collection from new model calls under different `model.version`, region, prompt-cache state, or collection wall-clock window could produce different traces. The trace-integrity rejection of one host’s slice (§4.9) is itself an instance of the same sensitivity — a multi-run trace concatenation produced visibly different distributional structure than the canonical single-run slices, and was correctly rejected by the pre-committed integrity guard before any analysis used it. Stage 3 instrumentation closes this gap.

6.6 Pre-registered exclusion criteria (none triggered)

For transparency: we pre-registered specific exclusion criteria for repetitions (parse-failure rate $> 10\%$ over a repetition; connectivity $< 50\%$ of design; collapsed-mode outputs in $> 95\%$ of rounds, where “collapsed” is defined by identical-string output across consecutive rounds). None of the three criteria were triggered in any Stage 2b repetition: parse-failure rate was 0.000% across all 9000 agent-round invocations, connectivity was 100%, and collapsed-mode outputs did not appear at any horizon. We did not have to make any post-hoc inclusion or exclusion decisions. This is reported here for procedural transparency, not as a substantive limitation.

6.7 Stage 2b sample-size envelope

Five Stage 2b results report a directionally-consistent point estimate whose 95% confidence interval includes the null. Stage 2b is positioned as a directional, pre-registration-locked stage; Stage 3 is the formal test. Table 8 consolidates the five results in one place; body sections (§4.4, §4.7, §4.8, §4.9, §5.5) cite this table once each rather than re-litigating the consistent-with-but-CI-spans-zero structure per result.

Table 8: Stage 2b sample-size envelope: five results whose point estimates are directionally consistent with the pre-registered hypothesis but whose 95% confidence intervals include the null. Stage 3 redress per Addendum #1 §F. *Multiplicity context* (cross-reference to §5.2.1): the SR-M-1...7 registry is sequentially registered, not formally FWER-corrected; Bonferroni-7 ≈ 0.0071 if family-wise control across the seven mechanistic claims is desired.

Result	n_{rep}	Point estimate	95% CI / p	Stage 3 redress
Basin-stability at $n = 2$ (§4.4)	15	12/15 = 0.80	Wilson [0.548, 0.930] overlaps 0.80 threshold	$n_{\text{rep}} \geq 30$, horizon 100
Reasoning- Δ_{profit} at $n = 5$ (§4.7)	30	$r = -0.30$	Fisher- z $[-0.60, +0.06]$	cross-model replication
Cluster-mean reasoning effect, GATE-5 $k = 2$ (§4.8)	30	$d = 0.60$	BCa $[-6.6, +66.7]$ chars	cross-model replication
Host effect at GATE-2 (§4.9)	15 (10/5)	OR = 13.5	Fisher’s $p = 0.0769$	multi-rep host audit, $n_{\text{rep}} \geq 30$
Wu 2025 simplicity-bias at $n = 2$ (§5.5)	15	$r = -0.47$	Fisher- z $[-0.79, +0.05]$	cross-model replication

The Stage 3 expansion to $n_{\text{rep}} \geq 30$ at horizon 100 is pre-committed (Addendum #1 §F) and is structured to discriminate each of the five results above. We note that the only post-hoc power statistic in the paper (§4.7, 39% at $r = -0.30$) is reported once in body and not duplicated here, since the table reports observed CIs rather than computed power.

7 Conclusion and Future Work

We report Stage 2b of a pre-registered pilot study of LLM-agent Bertrand pricing dynamics, with a headline finding of comparable terminal profit gain ($\Delta_{\text{profit}} = 0.4932$ at $n = 5$ versus 0.4522 at $n = 2$, both capturing approximately half the Nash-to-Monopoly profit gap) coupled to sharply divergent convergence rates (30/30 at $n = 5$ versus 1/15 at $n = 2$ within a 50-round horizon). The asymmetry is consistent with a wider basin of attraction at smaller agent counts and is the central empirical contribution of this paper.

The methodological contribution is the layered integrity stack: cryptographic pre-registration on OSF Registries with RFC 3161 trusted timestamps as the foundation, independent cross-toolchain reproduction (TypeScript primary analyzer plus Python/SciPy reference verifier) as a procedural reinforcing layer, and adversarial multi-persona expert-panel review with RedTeam adversarial calibration as an iterative reinforcing layer. The three commitments are not equivalent: cryptographic pre-registration is the strongest integrity primitive available for analysis-plan locking; the verification and adversarial layers reinforce it but cannot substitute for it. The stack caught one analyzer bug pre-submission (a Fisher’s exact denominator error that would have inverted the host-effect verdict); fired one pre-committed downgrade branch literally on the data ($k = 4$ to $k = 2$ at GATE-5); and produced six required revisions to the analysis prior to this preprint, three of which materially tightened the published claim wording. The protocol is replicable: every layer is open-source and the procedure generalizes to any pre-registered multi-agent LLM study.

Stage 3 expands n_{rep} to ≥ 30 for both conditions and lengthens the horizon to 100 rounds, and instruments the trace schema with `model_version`, `region`, `ttft_ms`, `tpot_ms`, `cache_hit_tokens`, `cache_miss_tokens`, and `auth_handshake_ms` per Addendum #1 §F to isolate LLM-serving variability from reasoning-depth variability. Three falsification conditions are pre-locked in this paper (§5.4): the basin-width-by-agent-count hypothesis, the Δ_{profit} asymmetry magnitude, and the LLM-versus- Q -learner family-generalization claim. Stage 4 conducts cross-model replication across Haiku sizes, Sonnet, Opus, and non-Anthropic frontier models to formally test the simplicity-bias mechanism (Wu et al., 2025; §5.5) and the family-generalization claim.

The codebase, the pre-registered analysis pipeline, the four RFC 3161-stamped addenda, and the full Verification & Correction Log are publicly available at <https://github.com/HHA-Applied-Research-Institute/colludebench-cascade>. We invite Stage 3 replication — particularly at the duopoly limit where the Stage 2b power envelope is thinnest — and the parallel Q -learner positive-control experiment pre-committed in `pilot/VALIDATION_PROTOCOL.md` as the cross-family discriminator that this pilot defers to.

References

- Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models, 2023. URL <https://arxiv.org/abs/2305.16867>.
- Galit Askenazi-Golan, Domenico Mergoni Cecchelli, Edward Plumb, and Clemens Possnig. The

- bounds of algorithmic collusion: Q-learning, gradient learning, and the folk theorem, 2024. URL <https://arxiv.org/abs/2411.12725>.
- John Asker, Chaim Fershtman, and Ariel Pakes. Artificial intelligence and pricing: The impact of algorithm design. Technical report, 2021. URL <https://www.nber.org/papers/w28535>.
- Stephanie Assad, Robert Clark, Daniel Ershov, and Lei Xu. Algorithmic pricing and competition: Empirical evidence from the german retail gasoline market. *Journal of Political Economy*, Vol. 132, No. 3, pp. 723-771, 2024. URL <https://doi.org/10.1086/726906>.
- Michel Benaim, Josef Hofbauer, and Sylvain Sorin. Stochastic approximations and differential inclusions. *SIAM Journal on Control and Optimization*, Vol. 44, No. 1, pp. 328-348, 2005. URL <https://epubs.siam.org/doi/10.1137/S0363012904439301>.
- Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024. URL <https://arxiv.org/abs/2407.21787>.
- Z.Y. Brown and A. MacKay. Competition in pricing algorithms. *American Economic Journal: Microeconomics* 15(2), 2023. URL <https://www.aeaweb.org/articles?id=10.1257/mic.20210158>.
- Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolo, and Sergio Pastorello. Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, Vol. 110, No. 10, pp. 3267-3297, 2020. URL <https://www.aeaweb.org/articles?id=10.1257/aer.20190623>.
- S. Cao and M. Hu. Llm collusion, 2026. URL <https://arxiv.org/abs/2601.01279>.
- Ariel Ezrachi and Maurice E. Stucke. Virtual competition: The promise and perils of the algorithm-driven economy. *Harvard University Press*, 2016. URL <https://www.hup.harvard.edu/books/9780674545472>.
- Sara Fish, Yannai A. Gonczarowski, and Ran I. Shorrer. Algorithmic collusion by large language models, 2024. URL <https://arxiv.org/abs/2404.00806>.
- Drew Fudenberg and David K. Levine. The theory of learning in games, 1998. URL <https://mitpress.mit.edu/9780262529242/the-theory-of-learning-in-games/>.
- L. Hammond, A. Chan, J. Clifton, and et al. Multi-agent risks from advanced ai, 2025. URL <https://arxiv.org/abs/2502.14143>.
- John C. Harsanyi and Reinhard Selten. A general theory of equilibrium selection in games, 1988. URL <https://mitpress.mit.edu/9780262581127/a-general-theory-of-equilibrium-selection-in-games/>.
- Connor Holmes, Masahiro Tanaka, Michael Wyatt, Ammar Ahmad Awan, Jeff Rasley, Samyam Rajbhandari, Reza Yazdani Aminabadi, Heyang Qin, Arash Bakhtiari, Lev Kurilenko, and Yuxiong He. Deepspeed-fastgen: High-throughput text generation for llms via mii and deepspeed-inference, 2024. URL <https://arxiv.org/abs/2401.08671>.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention, 2023. URL <https://arxiv.org/abs/2309.06180>.

- Pratyush Patel, Esha Choukse, Chaojie Zhang, Aashaka Shah, Íñigo Goiri, Saeed Maleki, and Ricardo Bianchini. Splitwise: Efficient generative llm inference using phase splitting, 2023. URL <https://arxiv.org/abs/2311.18677>.
- C. Riedl. Emergent coordination in multi-agent language models, 2025. URL <https://arxiv.org/abs/2510.05174>.
- William H. Sandholm. Population games and evolutionary dynamics, 2010. URL <https://mitpress.mit.edu/9780262195874/population-games-and-evolutionary-dynamics/>.
- Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters, 2024. URL <https://arxiv.org/abs/2408.03314>.
- George J. Stigler. A theory of oligopoly. *Journal of Political Economy*, 72(1):44–61, 1964. URL <https://doi.org/10.1086/258853>.
- O. Tailor. Audit the whisper: Detecting steganographic collusion in multi-agent llms, 2025. URL <https://arxiv.org/abs/2510.04303>.
- Robert Tibshirani, Guenther Walther, and Trevor Hastie. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(2):411–423, 2001. URL <https://doi.org/10.1111/1467-9868.00293>.
- Ludo Waltman and Uzay Kaymak. Q-learning agents in a cournot oligopoly model. *Journal of Economic Dynamics and Control*, Vol. 32, No. 10, pp. 3275–3293, 2008. URL <https://doi.org/10.1016/j.jedc.2008.01.003>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS 2022 (Advances in Neural Information Processing Systems)*, 2022. URL <https://arxiv.org/abs/2201.11903>.
- Yuyang Wu, Yifei Wang, Ziyu Ye, Tianqi Du, Stefanie Jegelka, and Yisen Wang. When more is less: Understanding chain-of-thought length in llms, 2025. URL <https://arxiv.org/abs/2502.07266>.
- Michael Wunder, Michael L. Littman, and Monica Babes. Classes of multiagent q-learning dynamics with epsilon-greedy exploration, 2010. URL <https://icml.cc/Conferences/2010/papers/191.pdf>.
- H. Peyton Young. The evolution of conventions. *Econometrica*, Vol. 61, No. 1, pp. 57–84, 1993. URL <https://www.jstor.org/stable/2951778>.
- Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. Sglang: Efficient execution of structured language model programs, 2023. URL <https://arxiv.org/abs/2312.07104>.